Chapter 7

Newton's Method in Infinite Dimensions

In Chapter 2 we studied the theory of solving $f(x) = 0$ where f is defined on a finite dimensional vector space. We now direct our attention to solving $f(x) = 0$ where x is an unknown function. This leads us to consider infinite dimensional vector spaces, known as Banach spaces.

This chapter presents some general background material on Banach spaces. We briefly study linear operators between Banach spaces and extends some of the basic notions of differential calculus to this new setting. We consider $f(x) = 0$ problems for differentiable maps between Banach spaces and extend the earlier notions of Newton-like operators and radii-polynomials. This material is the basis of the computer assisted proofs on function spaces developed in the remainder of the book.

7.1 Banach Spaces and Bounded Linear Operators

Definition 7.1.1. A *Banach space* is a normed vector space that is complete in the metric defined by the norm.

The finite dimensional vector spaces $\mathbb{R}^n$, $\mathbb{C}^n$, $M(\mathbb{R}^n, \mathbb{R}^m)$, and $M(\mathbb{C}^n, \mathbb{C}^m)$ are Banach spaces. A variety of Banach spaces are introduced in this chapter, but as the following example indicates not every linear function space is a Banach space.

Example 7.1.2. Consider $C^0([0,1], \mathbb{R})$ with the norm

$$\|f\|_1 = \int_0^1 |f(t)|\, dt. \tag{7.1}$$

For $n \geq 1$ define the sequence of continuous functions

$$f_n(x) := \begin{cases} 1 - nx & \text{if } 0 \leq x \leq 1/n, \\ 0 & \text{if } 1/n \leq x \leq 1. \end{cases}$$

Observe that $\{f_n\}$ is a Cauchy sequence in the norm defined by (7.1). However, $\{f_n\}$ does not converge pointwise to a continuous function and thus we do not have a Banach space.

One consequence of Example 7.1.2 is that to identify a Banach space it is essential to fix both the vector space and the norm. For the simplicity, in the abstract setting if we state that V is a Banach space, then we denote the associated norm by $\|\cdot\|_V$. In settings where the norm is clear we will simplify further and write $\|\cdot\|$.

Given a finite collection of Banach spaces $V_1, \ldots, V_L$, the cartesian product can be made into a Banach space. More precisely let

$$V := V_1 \oplus \ldots \oplus V_L = \{(v_1, \ldots, v_L) \mid v_l \in V_l \text{ for } 1 \leq l \leq L\}$$

and for $v = (v_1, \ldots, v_L) \in V$ define

$$\|v\|_V := \max_{1 \leq l \leq L} \|v_l\|_{V_l}.$$

It is left to the reader to check that V is a Banach space under the norm $\|\cdot\|_V$.

Given $V = V_1 \oplus \ldots \oplus V_L$ the *canonical projection* operators, $\pi^l \colon V \to V_l$ $l = 1, \ldots, L$, are defined by

$$\pi^l(v_1, \ldots, v_L) = v_l.$$

Definition 7.1.3. Let V and W be normed vector spaces. A linear operator $A \colon V \to W$ is *bounded* if

$$\|A\| = \|A\|_{B(V,W)} := \sup_{\|v\|_V = 1} \|Av\|_W < \infty.$$

The space of all bounded linear operators from V to W is denoted by $B(V,W)$. If $V = W$, then we simplify the notation and write $B(V)$.

It is left to the reader to check that if V and W are finite dimensional and $A \colon V \to W$ is a linear map, then $A \in B(V,W)$.

Proposition 7.1.4. *Let $V = V_1 \oplus \ldots \oplus V_L$ and let $\pi^l \colon V \to V_l$ be the canonical projection operator. Then, $\|\pi^l\|_{B(V,V_l)} = 1$.*

Proof. Let $v \in V$ such that $\|v\|_V = 1$. Observe that

$$\sup_{\|v\|_V = 1} \|\pi^l v\|_{V_l} = \sup_{\|v_l\|_V = 1} \|v_l\|_{V_l} = 1.$$

□

Proposition 7.1.5. *Let V be a normed vector space. Let $I: V \to V$ be the identity map and let $M: V \to V$ be a linear operator. If $\|I - M\|_{B(V)} < 1$, then A is injective.*

Proof. Assume that A is not injective. Then, there exists $v \in V$ such that $Av = 0$. Let $w = v/\|v\|$. Observe that $(I - A)w = w$. Therefore, $\|I - A\|_{B(V)} \geq 1$, a contradiction. □

Example 7.1.6. Consider the space $C^0([a, b])$ with the supremum norm. Define the linear operator $A: C^0([a, b]) \to C^0([a, b])$ by

$$(Af)(t) = \int_a^t f(x)\, dx.$$

By the fundamental theorem of calculus A is well defined, i.e., Af is a continuous function. The following calculation shows that A is a bounded.

$$\begin{aligned}
\|A\| &= \sup_{\|f\|=1} \|Af\|_\infty \\
&= \sup_{\|f\|=1} \sup_{t\in[a,b]} |(Af)(t)| \\
&= \sup_{\|f\|=1} \sup_{t\in[a,b]} \left| \int_a^t f(x)\, dx \right| \\
&\leq \sup_{\|f\|=1} \sup_{t\in[a,b]} \int_a^t |f(x)| dx \\
&\leq \sup_{\|f\|=1} \sup_{t\in[a,b]} \int_a^t \sup_{y\in[a,b]} |f(y)| dx \\
&\leq \sup_{\|f\|=1} \int_a^b \|f\|_\infty dx \\
&\leq \sup_{\|f\|=1} (b - a)\|f\|_\infty \\
&\leq b - a.
\end{aligned}$$

The operator norm gives rise to the following important inequality.

Proposition 7.1.7. *If $A \in B(V, W)$, then $\|Ax\|_W \leq \|A\|_{B(V,W)} \|x\|_V$ for all $x \in V$.*

Proof. Let $x \in V$. If $x = 0$ then $Ax = 0$ and we have the equality

$$\|Ax\|_W = \|0\|_W = 0 = \|A\|\|x\|_V.$$

If $x \neq 0$, then $0 < \|x\|_V$ and

$$\begin{aligned}\|Ax\|_W &= \left\| Ax \frac{\|x\|_V}{\|x\|_V} \right\|_W \\ &= \|x\|_V \left\| A \frac{x}{\|x\|_V} \right\|_W \\ &\leq \|x\|_V \sup_{\|y\|_V=1} \|Ay\|_W \\ &= \|A\| \|x\|_V.\end{aligned}$$

□

A similar argument gives rise to the following proposition.

Proposition 7.1.8. *Let V be a normed linear space. If $A, B \in B(V)$, then*

$$\|AB\| \leq \|A\| \|B\|.$$

Part of the importance of bounded linear operators is that they characterize continuous linear operators.

Theorem 7.1.9. *Let V and W be normed linear spaces and $A \colon V \to W$ be a linear operator. Then A is a continuous map from V to W if and only if $A \in B(V, W)$.*

Proof. Suppose that $A \in B(V, W)$. Given $x \in V$, we prove that A is continuous at x. Consider a sequence $\{x_n\}_{n=0}^{\infty} \subset V$ that converges to x. By Proposition 7.1.7

$$\begin{aligned}\|Ax - Ax_n\|_W &= \|A(x - x_n)\|_W \\ &\leq \|A\| \|x - x_n\|_V\end{aligned}$$

and hence $\|Ax - Ax_n\|_W \to 0$ as $n \to \infty$, i.e.

$$\lim_{n \to \infty} Ax_n = Ax,$$

as desired.

Now suppose that $A \colon V \to W$ is continuous. Then A is continuous at $0 \in V$, and hence there exists $\delta > 0$ such that $\|x\|_V \leq \delta$ implies that $\|Ax\|_W \leq 1$. Let $y \in V$ such that $\|y\|_V = 1$. Then $\|\delta y\| = \delta$ and

$$\|Ay\|_W = \frac{1}{\delta} \|A(\delta y)\|_W \leq \frac{1}{\delta}.$$

Since this bound is uniform in y

$$\|A\| = \sup_{\|y\|_V=1} \|Ay\|_W \leq \frac{1}{\delta}$$

and hence $A \in B(V, W)$. □

It is important to understand when $B(V, W)$ is itself a Banach space.

Theorem 7.1.10. *Let V be a normed vector space. If W is a Banach space, then $B(V, W)$ is a Banach space.*

Proof. Let $\{A_n\}_{n\in\mathbb{N}} \subset B(V, W)$ be a Cauchy sequence of bounded linear operators.

We begin by proving that there exists a function $A\colon V \to W$ defined by

$$Av \stackrel{\text{def}}{=} \lim_{n\to\infty} A_n v, \quad v \in V.$$

Since $\{A_n\}$ is Cauchy, given $\epsilon > 0$, there exists $N > 0$ such that $\|A_n - A_m\| < \epsilon$, for any $n, m > N$ and hence

$$\|A_n v - A_m v\|_W < \|A\|\|v\|_V < \epsilon\|v\|_V,$$

for all $n, m > N$. Thus, for each $v \in V$, $\{A_n v\}$ is Cauchy. Since W is a Banach space, $\lim_{n\to\infty} A_n v$ converges.

We now prove that $A \in B(V, W)$. We begin by showing that A is a linear operator. Given $v \in V$, a scalar α and $\epsilon > 0$, choose n sufficiently large such that

$$\|Av - A_n v\|_W < \epsilon \quad \text{and} \quad \|A\alpha v - A_n \alpha v\|_W < \epsilon.$$

Then

$$\begin{aligned}
\|A\alpha v - \alpha Av\|_W &\leq \|A\alpha v - A_n\alpha v\|_W + \|A_n\alpha v - \alpha Av\|_W \\
&\leq \epsilon + \|\alpha A_n v - \alpha Av\|_W \\
&\leq \epsilon(1 + |\alpha|).
\end{aligned}$$

Since this is true for all ϵ, $A\alpha v = \alpha Av$. A similar argument shows that $A(v+v') = Av+Av'$.

We need to show that A is bounded. Using the fact that $\{A_n\}$ is Cauchy there exists $n > 0$ such that if $m \geq n$, then

$$\|A_n - A_m\| < 1.$$

Choose $v \in V$ such that $\|v\|_V = 1$ and observe that

$$\|A_m v\|_W \leq \|A_m v - A_n v\|_W + \|A_n v\|_W \leq 1 + \|A_n\|.$$

Thus,

$$\|A\| \leq \limsup_m \|A_m\| \leq 1 + \|A_n\|$$

and hence A is bounded. □

Definition 7.1.11. A linear operator $A\colon V \to W$ is *boundedly invertible* if A is invertible and

$$\|A^{-1}\| < \infty.$$

Note that in the definition above it is not necessary that A itself is bounded. Indeed, there are important applications where an unbounded linear operator has a bounded inverse. An illustrative example is differentiation of analytic functions which in general does not lead to a bounded operator (see Remark 8.4.10), but its inverse integration does, e.g. Example 7.1.6. That being said, if A is bounded we have the following theorem.

Theorem 7.1.12 (Bounded Inverse Theorem). *Suppose that $A\colon V \to W$ is a bijective linear map with V and W Banach spaces. If $A \in B(V,W)$ then $A^{-1} \in B(W,V)$, i.e. if A is bounded and A^{-1} exists, then the inverse is bounded as well.*

An important special case of linear maps from V to W arises when $W = \mathbb{R}$ (or $W = \mathbb{C}$ if V is a vector space over $\mathbb{C}$). In this case the linear map is called a *linear functional* on V.

Definition 7.1.13. The set of all bounded linear functionals on V is called the *dual space* of V and is denoted by V^*.

By Theorem 7.1.10 V^* is a Banach space.

Example 7.1.14. Consider the linear functional $\ell\colon C^0([a,b]) \to \mathbb{R}$ defined by

$$\ell(f) = \int_a^b f(x)\,dx.$$

It is an exercise to show that $\ell \in C^0([a,b])^*$, i.e., that ℓ is bounded.

We conclude this section with a discussion concerning invertible elements of $B(V)$.

Theorem 7.1.15 (The Neumann Series). *Let V be a Banach space and $A \in B(V)$. Let I denote the identity map on V. If $\|A\| < 1$, then*

(i) *$I - A$ is invertible,*

(ii)

$$(I-A)^{-1} = \sum_{n=0}^{\infty} A^n,$$

(iii)

$$\|(I-A)^{-1}\| \leq \frac{1}{1-\|A\|}.$$

Proof. First note that, since we assume that V is a Banach space, the space $B(V)$ is also a Banach space by Theorem 7.1.10. Define the sequence of partial sums $\{B_n\}_{n=0}^{\infty} \subset B(V)$

by $B_n = \sum_{k=0}^{n} A^k$. We will show that the sequence $\{B_n\}_{n=0}^{\infty}$ is Cauchy. To see this let $\epsilon > 0$ be given. Choose $N \in \mathbb{N}$ so that

$$\frac{\|A\|^{N+1}}{1 - \|A\|} < \epsilon.$$

Note that this N exist by the hypothesis that $\|A\| < 1$. Now for $m, n \geq N$ we assume without loss of generality that $n > m$ and have that

$$B_n - B_m = \sum_{k=0}^{n} A^k - \sum_{k=0}^{m} A^k = \sum_{k=m+1}^{n} A^k.$$

Then

$$\begin{aligned}
\|B_n - B_m\| &= \left\| \sum_{k=m+1}^{n} A^k \right\| \\
&\leq \sum_{k=m+1}^{n} \|A^k\| \\
&\leq \sum_{k=m+1}^{n} \|A\|^k \\
&\leq \sum_{k=m+1}^{\infty} \|A\|^k \\
&\leq \sum_{k=N}^{\infty} \|A\|^k \\
&\leq \frac{\|A\|^{N+1}}{1 - \|A\|} \\
&< \epsilon,
\end{aligned}$$

and indeed the sequence is Cauchy. Since $B(V)$ is complete, there exists a $B \in B(V)$ to which the partial sums converge, i.e.

$$B = \lim_{N \to \infty} B_N = \lim_{N \to \infty} \sum_{k=0}^{N} A^k = \sum_{k=0}^{\infty} A^k.$$

We now claim that B inverts $I - A$. To see this note first that $\|A^n\| \leq \|A\|^n \to 0$ as $n \to \infty$, so that $A^n \to 0$ as $n \to \infty$. Now observe that

$$(I - A)B_n = B_n - AB_n = \sum_{k=0}^{n} A^k - \sum_{k=1}^{n+1} A^k = I - A^{n+1},$$

and similarly that

$$B_n(I-A) = B_n - B_nA = \sum_{k=0}^{n} A^k - \sum_{k=1}^{n+1} A^k = I - A^{n+1}.$$

Taking limits yields $(I-A)B = I$ and $B(I-A) = I$. So $(I-A)$ is invertible with inverse $B = \sum_{n=0}^{\infty} A^n$.

Finally we note that

$$\|(I-A)^{-1}\| = \left\| \sum_{n=0}^{\infty} A^n \right\| \le \sum_{n=0}^{\infty} \|A^n\| \le \sum_{n=0}^{\infty} \|A\|^n = \frac{1}{1-\|A\|},$$

by applying the geometric series for the *number* $\|A\| < 1$, yielding the desired estimate. □

7.2 Calculus on Banach Spaces

Definition 7.2.1. Let V, W be Banach spaces. A function $F\colon V \to W$ is *Fréchet differentiable* at $x_0 \in V$ if there exists a bounded linear operator $A\colon V \to W$ satisfying the following condition: given any $\epsilon > 0$, there exists $\delta > 0$ such that if $\|h\| < \delta$, then

$$\|F(x_0 + h) - F(x_0) - Ah\|_W \le \epsilon \|h\|_V.$$

The linear operator A is called the *Fréchet derivative* of F at x_0 and denoted by $A = DF(x_0)$.

A convenient short hand for expressing this relation is to write

$$\lim_{h \to 0} \frac{\|F(x_0+h) - F(x_0) - Ah\|_W}{\|h\|_V} = 0.$$

Example 7.2.2. Consider the Banach space $C^0([a,b],\mathbb{R})$ with the sup norm and define the mapping $F\colon C^0([a,b],\mathbb{R}) \to C^0([a,b],\mathbb{R})$ by

$$F[u](t) = \int_a^t u^2(x)\,dx.$$

We claim that for any $u_0 \in C^0([a,b],\mathbb{R})$ the derivative of F at u_0 is given by

$$DF(u_0)[h](t) = 2\int_a^t u_0(x)h(x)\,dx.$$

Thus we need to show that given $\epsilon > 0$, there exists $\delta > 0$ such that if $\|h\|_\infty < \delta$ then

$$\left\| F[u_0+h] - F[u_0] - 2\int_a^t u_0(x)h(x)\,dx \right\|_\infty < \epsilon \|h\|_\infty$$

To this end, observe that

$$\begin{aligned} F[u_0+h]-F[u_0] &= \int_a^t [u_0(x)+h(x)]^2\,dx - \int_a^t u_0^2(x)\,dx \\ &= \int_a^t \left(u_0^2(x)+2u_0(x)h(x)+h^2(x)\right)\,dx - \int_a^t u_0^2(x)\,dx \\ &= 2\int_a^t u_0(x)h(x)\,dx + \int_a^t h^2(x)\,dx \end{aligned}$$

and given $\epsilon > 0$ set $\delta = \epsilon/(b-a)$. If $\|h\|_\infty < \delta$, then

$$\left\| \int_a^t h^2(x)\,dx \right\| \le \int_a^t \|h\|^2\,dx \le (b-a)\|h\|^2 \le \epsilon\|h\|,$$

which is the desired inequality.

The proof of the following result is left as an Exercise.

Proposition 7.2.3. *Let X be a Banach space. If $A\colon X \to X$ is a bounded linear operator and $F\colon X \to X$ is Fréchet differentiable at $x_0 \in X$, then the mapping $G\colon X \to X$ defined by $G(x) = AF(x)$ is Fréchet differentiable at x_0 and $DG(x_0) = ADF(x_0)$.*

The following two fundamental results from calculus generalize to the setting of Banach spaces.

Theorem 7.2.4 (Mean Value Theorem). *Let V, W be Banach spaces. Let $u_0 \in V$ and suppose that $F\colon \overline{B_r(u_0)} \subset V \to W$ is a C^1 mapping, i.e. we assume that for every $v \in \overline{B_r(u_0)}$ the Fréchet derivative $DF(v)$ exists and is a bounded linear operator. Let*

$$K \stackrel{\text{def}}{=} \sup_{v \in \overline{B_r(u_0)}} \|DF(v)\|.$$

Then for any $u, v \in \overline{B_r(u_0)}$ we have that

$$\|F(u) - F(v)\|_W \le K\|u - v\|_V.$$

Proof. See the proofs of Theorem 1.225 and Theorem 1.226 in [10]. □

Theorem 7.2.5 (Implicit Function Theorem). *Let X, Y, and Z denote Banach spaces. Assume $U \subset X$, $V \subset Y$ are open sets, $F\colon U \times V \to Z$ is C^1, and $(x_0, y_0) \in U \times V$ with $F(x_0, y_0) = 0$. If $D_xF(x_0, y_0)\colon X \to Z$ has a bounded inverse, then there is a neighborhood $U_0 \times V_0 \subset U \times V$ of (x_0, y_0) and a C^1 function $\alpha\colon V_0 \to U_0$ such that*

$$F(\alpha(y), y) = 0 \quad \text{and} \quad \alpha(y_0) = x_0.$$

Furthermore, if $(x, y) \in U_0 \times V_0$ and $F(x, y) = 0$, then $x = \alpha(y)$.

7.3 Banach Spaces of Infinite Sequences

While there are a wide variety of Banach spaces the goals of this book lead us to concentrate on spaces whose elements consist of infinite sequences. The essential idea is that numerical approximations derived via computations consist of a finite set of numbers (the first terms in the sequence) and to obtain a rigorous result requires bounds on the remaining elements of the sequence. A classical example, described more explicitly in Example 7.3.3 is that of Taylor series approximations of analytic functions.

The space of complex valued *one-sided sequences* (*two-sided sequences*) is

$$S_{\mathbb{I}}(\mathbb{C}) := \{a = \{a_n\} \mid a_n \in \mathbb{C}, n \in \mathbb{I}\}$$

where $\mathbb{I} = \mathbb{N}$ ($\mathbb{I} = \mathbb{Z}$). The space of real valued sequences $S_{\mathbb{I}}(\mathbb{R})$ is defined similarly. Observe that $S_{\mathbb{I}}(\mathbb{C})$ ($S_{\mathbb{I}}(\mathbb{R})$) is a vector space over $\mathbb{C}$ ($\mathbb{R}$) under component-wise addition and scalar multiplication.

A *sequence of weights over* $\mathbb{I}$ is a sequence $\omega = \{\omega_n\}_{n\in\mathbb{I}} \subset \mathbb{R}$ of positive numbers. Given a sequence of weights ω and an element $a \in S_{\mathbb{I}}(\mathbb{C})$ define

$$\|a\|_{1,\omega,\mathbb{I}} := \sum_{n\in\mathbb{I}} |a_n|\omega_n \quad \text{and} \quad \|a\|_{\infty,\omega,\mathbb{I}} := \sup_{n\in\mathbb{I}} |a_n|\omega_n.$$

The proof of the following lemma is left as an exercise.

Lemma 7.3.1. *If ω is a sequence of weights over $\mathbb{I}$, then for $\|\cdot\| \in \{\|\cdot\|_{1,\omega,\mathbb{I}}\ ,\ \|\cdot\|_{\infty,\omega,\mathbb{I}}\}$,*

(i) $\|a\| = 0$ if and only if $a_n = 0$ for all $n \in \mathbb{I}$;

(ii) if $\alpha \in \mathbb{C}$, then
$$\|\alpha a\| = |\alpha|\|a\|;$$

(iii) and,
$$\|a + b\| \leq \|a\| + \|b\|,$$
for all $a, b \in S_{\mathbb{I}}(\mathbb{C})$,

where by convention $0 \cdot \infty = 0$, $x \cdot \infty = \infty$ for $x \in (0, \infty)$, and $x + \infty = \infty$ for $x \in [0, \infty]$.

The space $S_{\mathbb{I}}(\mathbb{C})$ is too large. In particular, given a sequence of weights ω there exist elements $a \in S_{\mathbb{I}}(\mathbb{C})$ such that $\|a\| = \infty$. However, as the following theorem indicates, by restricting to elements for which $\|a\|$ is finite, we obtain Banach spaces.

Theorem 7.3.2. *If ω is a sequence of weights over $\mathbb{I}$, then*

$$\ell^1_{\omega,\mathbb{I}} := \{a \in S_{\mathbb{I}}(\mathbb{C}) \mid \|a\|_{1,\omega,\mathbb{I}} < \infty\}$$

and

$$\ell^\infty_{\omega,\mathbb{I}} := \{a \in S_{\mathbb{I}}(\mathbb{C}) \mid \|a\|_{\infty,\omega,\mathbb{I}} < \infty\}$$

are Banach spaces.

Proof. We present the proof for $\ell^1_{\omega,\mathbb{N}}$ and leave the remaining cases as exercises.

Since $a \in \ell^1_{\omega,\mathbb{N}}$ implies that $\|a\|_{1,\omega,\mathbb{N}} < \infty$, Lemma 7.3.1 implies that $\|\cdot\|_{1,\omega,\mathbb{N}}$ is a norm on $\ell^1_{\omega,\mathbb{N}}$. Furthermore, Lemma 7.3.1(ii) and (iii) implies that $\ell^1_{\omega,\mathbb{N}} \subset S_{\mathbb{N}}(\mathbb{C})$ is closed under addition and scalar multiplication. Therefore, $\ell^1_{\omega,\mathbb{N}}$ is a vector subspace of $S_{\mathbb{N}}(\mathbb{C})$. It remains to show that under the $\|\cdot\|_{1,\omega,\mathbb{N}}$ norm, $\ell^1_{\omega,\mathbb{N}}$ is a complete metric space.

To this end we prove that any Cauchy sequence

$$\left\{a^k = \left\{a^k_n\right\}_{n=0}^{\infty} \mid k = 0, 1, \ldots\right\} \subset \ell^1_{\omega,\mathbb{N}}$$

has a limit in $\ell^1_{\omega,\mathbb{N}}$. The proof has three steps. First, we construct a candidate $\hat{a} \in S_{\mathbb{N}}(\mathbb{C})$. Next, we prove that $\hat{a} \in \ell^1_{\omega,\mathbb{N}}$. Finally, we show that $\lim_{k\to\infty} a^k = \hat{a}$.

We begin by showing that for every $n \in \mathbb{N}$, $\left\{a^k_n\right\}$ is a Cauchy sequence. Fix $n \in \mathbb{N}$. Then,

$$\left|a^k_n\omega_n - a^j_n\omega_n\right| = \left|a^k_n - a^j_n\right|\omega_n \leq \sum_{n=0}^{\infty}\left|a^k_n - a^j_n\right|\omega_n = \|a^k - a^j\|_{1,\omega,\mathbb{N}}.$$

Thus $\left\{a^k_n\omega_n\right\}$ is Cauchy . Since $\omega_n \neq 0$ is a constant, $\left\{a^k_n\right\}$ is Cauchy and thus converges. Set $\hat{a} = \{\hat{a}_n\}_{n=0}^{\infty} \in S_{\mathbb{N}}(\mathbb{C})$ where

$$\hat{a}_n := \lim_{k\to\infty} a^k_n.$$

We now show that $\hat{a} \in \ell^1_{\omega,\mathbb{N}}$. Since $\left\{a^k\right\}$ is Cauchy, there is a $M \in \mathbb{N}$ so that for all $k, k' \geq K$

$$\|a^k - a^{k'}\|_{1,\omega,\mathbb{N}} < 1.$$

Choose $k \geq K$ and fix $N \in \mathbb{N}$. Then,

$$\begin{aligned}
\sum_{n=0}^{N}|a^k_n|\omega_n &\leq \sum_{n=0}^{N}|a^k_n - a^K_n|\omega_n + \sum_{n=0}^{N}|a^K_n|\omega_n \\
&\leq \|a^k - a^K\|_{1,\omega,\mathbb{N}} + \|a^K\|_{1,\omega,\mathbb{N}} \\
&\leq 1 + \|a^K\|_{1,\omega,\mathbb{N}}.
\end{aligned}$$

Observe that this estimate is independent of k. Thus,

$$S_N := \lim_{k\to\infty}\sum_{n=0}^{N}|a^k_n|\omega_n = \sum_{n=0}^{N}\lim_{k\to\infty}|a^k_n|\omega_n = \sum_{n=0}^{N}|\hat{a}_n|\omega_n \leq 1 + \|a^K\|_{1,\omega,\mathbb{N}},$$

and hence $\hat{a} \in \ell^1_{\omega,\mathbb{N}}$. Observe that $\{S_N\}_{N=0}^{\infty}$ is a monotone increasing sequence bounded by $\|a^K\|_{1,\omega,\mathbb{N}}$, and therefore converges. Thus,

$$\|a^K\|_{1,\omega,\mathbb{N}} = \lim_{N\to\infty} S_N < \infty$$

and hence $\hat{a} \in \ell^1_{\omega,\mathbb{N}}$ as desired.

The last step in the proof is to show that the Cauchy sequence $\{a^k\}$ actually converges to $\hat{a}$. Fix $\epsilon > 0$. Choose $K > 0$ such that $\|a^k - a^{k'}\| < \epsilon$ for all $k, k' \geq K$. Then, we obtain the bound

$$S_{N,k'} := \sum_{n=0}^{N} \left| a_n^k - a_n^{k'} \right| \omega_n \leq \|a^k - a^{k'}\|_{1,\omega,\mathbb{N}} < \epsilon.$$

This bound is independent of N and k'. Therefore,

$$\begin{aligned}
S_N := \lim_{k'\to\infty} S_{N,k'} &= \lim_{k'\to\infty} \sum_{n=0}^{N} \left| a_n^k - a_n^{k'} \right| \omega_n \\
&= \sum_{n=0}^{N} \lim_{k'\to\infty} \left| a_n^k - a_n^{k'} \right| \omega_n \\
&= \sum_{n=0}^{N} \left| a_n^k - \lim_{k'\to\infty} a_n^{k'} \right| \omega_n \\
&= \sum_{n=0}^{N} \left| a_n^k - \hat{a}_n \right| \omega_n \\
&\leq \epsilon,
\end{aligned}$$

and hence

$$\|a^k - \hat{a}\|_{1,\omega,\mathbb{N}} = \lim_{N\to\infty} \sum_{n=0}^{N} \left| a_n^k - \hat{a}_n \right| \omega_n < \epsilon.$$

This last bound is true for any ϵ, if k is sufficiently large, and therefore $\lim_{k\to\infty} a^k = \hat{a}$. $\square$

The construction of the weighted sequence spaces presented above is quite general. For the purposes of this book we are primarily interested in the following special cases.

Example 7.3.3. In Chapter 8 we study initial value problems for differential equations with polynomial nonlinearities using approximations by Taylor series and in Chapter 9 we use Taylor polynomials to describe nonlinear stable and unstable manifolds of equilibria.

Consider for the moment a scalar IVP $\dot{x} = f(x)$, $x(0) = x_0$ where the vector field $f\colon \mathbb{R} \to \mathbb{R}$ is a polynomial. Since f is analytic, the solution is analytic. More precisely, there exists $\tau > 0$ such that the solution $x\colon (-\tau, \tau) \to \mathbb{R}$ can we written in terms of its Taylor series expansion

$$x(t) = \sum_{n=0}^{\infty} a_n t^n. \tag{7.2}$$

Furthermore, this Taylor series expansion converges absolutely and uniformly for all $t \in (-\tau, \tau)$, that is

$$\sum_{n=0}^{\infty} |a_n| \nu^n < \infty$$

for all $\nu \in [0, \tau)$. Observe, that this implies that $a = \{a_n\}_n \in \mathbb{N} \in \ell^1_{\nu,\mathbb{N}}$ for all $\nu \in [0, \tau)$.

Because of its importance and for the sake of notational simplicity, given $\nu > 0$ we set

$$\ell^1_{\nu,\mathbb{N}} := \ell^1_{\omega,\mathbb{N}} \quad \text{where } \omega = \{\omega_n := \nu^n\}_{n \in \mathbb{N}}.$$

Similarly, we use the notation

$$\ell^\infty_{\nu,\mathbb{N}} := \ell^\infty_{\omega,\mathbb{N}} \quad \text{where } \omega = \{\omega_n := \nu^n\}_{n \in \mathbb{N}}.$$

Example 7.3.4. In Chapter 10 we use Fourier series to approximate periodic orbits. In this case the Fourier coefficients are naturally indexed by the integers and the sequence weights that we are interested in take the form

$$\omega = \left\{\omega_n := \nu^{|n|}\right\}_{n \in \mathbb{Z}} \qquad \text{for fixed } \nu \geq 1.$$

As in previous example, for the sake of notational simplicity, given $\nu \geq 1$ we set

$$\ell^1_{\nu,\mathbb{Z}} := \ell^1_{\omega,\mathbb{Z}} \quad \text{where } \omega = \left\{\omega_n := \nu^{|n|}\right\}_{n \in \mathbb{Z}}$$

and

$$\ell^\infty_{\nu,\mathbb{Z}} := \ell^\infty_{\omega,\mathbb{Z}} \quad \text{where } \omega = \left\{\omega_n := \nu^{|n|}\right\}_{n \in \mathbb{Z}}.$$

Example 7.3.5. In Chapter 14 we analyze initial value problems and boundary value problems using Chebyshev polynomials. For the sake of notational simplicity, given $K \in \mathbb{N}$ we set

$$\ell^1_{K,\mathbb{Z}} := \ell^1_{\omega,\mathbb{Z}} \quad \text{where } \omega = \left\{\omega_n := |n|^K + 1\right\}_{n \in \mathbb{Z}}$$

and

$$\ell^\infty_{K,\mathbb{Z}} := \ell^\infty_{\omega,\mathbb{Z}} \quad \text{where } \omega = \left\{\omega_n := |n|^K + 1\right\}_{n \in \mathbb{Z}}.$$

Definition 7.3.6. Let V be a complex Banach space. A countable collection $\{v_n\}_{n=0}^\infty \subset V$ is called a *Schauder basis* for V if for each $v \in V$ there is a unique collection of complex numbers $\{a_n\}_{n=0}^\infty$ so that

$$v = \sum_{n=0}^{\infty} a_n v_n.$$

Definition 7.3.7. The *standard basis vectors* on $\ell^1_{\omega,\mathbb{I}}$ are $\left\{e^k \mid k \in \mathbb{I}\right\}$ where

$$e^k_n = \begin{cases} 1 & \text{if } n = k \\ 0 & \text{otherwise.} \end{cases},$$

The following result is left to the reader.

Theorem 7.3.8. *The standard basis vectors on* $\left\{e^k \in \ell^1_{\omega,\mathbb{I}} \mid k \in \mathbb{I}\right\}$ *is a Schauder basis. Furthermore,* $\|e^k\|_{1,\omega,\mathbb{I}} = \omega_k$.

Definition 7.3.9. Two Banach spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are *isomorphic* if there exists a linear bijection $J\colon X \to Y$ such that J and its inverse J^{-1} are continuous. We say that X and Y are *isometrically isomorphic* if in addition, J is an *isometry*, that is $\|J(x)\|_Y = \|x\|_X$ for every $x \in X$.

The following Theorem provides a useful characterization of the Banach space dual of a space of summable sequences. We state the theorem for one sided sequences and leave the generalization to two sided sequences as an exercise.

Theorem 7.3.10. *Let* $\omega = \{\omega_n\}_{n=0}^{\infty}$ *be an sequence of weights and set* $\omega^{-1} := \{\omega_n^{-1}\}_{n=0}^{\infty}$. *Then,* $\ell^\infty_{\omega^{-1},\mathbb{N}}$ *and* $(\ell^1_{\omega,\mathbb{N}})^*$ *are isometrically isomorphic.*

In less precise language, we say that $\ell^\infty_{\omega^{-1},\mathbb{N}}$ is the dual of $\ell^1_{\omega,\mathbb{N}}$. The following estimate is used in the proof of Theorem 7.3.10.

Lemma 7.3.11. *If* $a \in \ell^1_{\omega,\mathbb{N}}$ *and* $c \in \ell^\infty_{\omega^{-1},\mathbb{N}}$, *then*

$$\left|\sum_{n=0}^{\infty} c_n a_n\right| \le \|c\|_{\infty,\omega^{-1},\mathbb{N}} \|a\|_{1,\omega,\mathbb{N}}.$$

Proof. Given $c \in \ell^\infty_{\omega^{-1},\mathbb{N}}$ and $k \ge 0$ we have, by definition of the weighted supremum norm, that

$$\frac{|c_n|}{\omega_n} \le \|c\|_{\infty,\omega^{-1},\mathbb{N}}.$$

Therefore,

$$\begin{aligned}
\left|\sum_{n=0}^{\infty} c_n a_n\right| &\le \sum_{n=0}^{\infty} |c_n||a_n| \\
&\le \sum_{n=0}^{\infty} \left(\|c\|_{\infty,\omega^{-1},\mathbb{N}}\, \omega_n\right)\, |a_n| \\
&= \|c\|_{\infty,\omega^{-1},\mathbb{N}} \sum_{n=0}^{\infty} |a_n|\omega_n \\
&= \|c\|_{\infty,\omega^{-1},\mathbb{N}} \|a\|_{1,\omega,\mathbb{N}}.
\end{aligned}$$

$\square$

Proof of Theorem 7.3.10. We begin by constructing an invertible linear map $J\colon \ell^\infty_{\omega^{-1},\mathbb{N}} \to (\ell^1_{\omega,\mathbb{N}})^*$. Given an element $c \in \ell^\infty_{\omega^{-1},\mathbb{N}}$, define Q_c, a linear functional on $\ell^1_{\omega,\mathbb{N}}$, by

$$Q_c a \stackrel{\text{def}}{=} \sum_{n=0}^{\infty} c_n a_n.$$

Then set

$$Jc \stackrel{\text{def}}{=} Q_c.$$

We leave it as an exercise to check that J is an linear transformation.

To show that the image of J is in $(\ell^1_{\omega,\mathbb{N}})^*$ we need to prove that Q_c is bounded. To this end observe that

$$\begin{aligned}
\|Q_c\| &\stackrel{\text{def}}{=} \sup_{\|a\|_{1,\omega,\mathbb{N}}=1} |Q_c a| \\
&= \sup_{\|a\|_{1,\omega,\mathbb{N}}=1} \left| \sum_{n=0}^{\infty} c_n a_n \right| \\
&\leq \sup_{\|a\|_{1,\omega,\mathbb{N}}=1} \|c\|_{\infty,\omega^{-1},\mathbb{N}} \|a\|_{1,\omega,\mathbb{N}} && \text{by Lemma 7.3.11} \\
&= \|c\|_{\infty,\omega^{-1},\mathbb{N}}. && (7.3)
\end{aligned}$$

We now prove that J is bounded and hence by Theorem 7.1.9 that J is continuous.

$$\begin{aligned}
\|J\|_{B\left(\ell^\infty_{\omega^{-1},\mathbb{N}},(\ell^1_{\omega,\mathbb{N}})^*\right)} &= \sup_{\|c\|_{\infty,\omega^{-1},\mathbb{N}}=1} \|Jc\|_{(\ell^1_{\omega,\mathbb{N}})^*} \\
&= \sup_{\|c\|_{\infty,\omega^{-1},\mathbb{N}}=1} \|Q_c\| \\
&\leq \sup_{\|c\|_{\infty,\omega^{-1},\mathbb{N}}=1} \|c\|_{\infty,\omega^{-1},\mathbb{N}} && \text{by (7.3)} \\
&= 1, && (7.4)
\end{aligned}$$

To show that J is injective, we prove that the kernel of J is trivial. Assume that $Jc = 0$. This means that $(Jc)(a) = \sum_{n=0}^{\infty} a_n c_n = 0$ for all $a \in \ell^1_{\omega,\mathbb{N}}$. In particular, for each e^k in the standard basis of $\ell^1_{\omega,\mathbb{N}}$,

$$(Jc)(e^k) = \sum_{n=0}^{\infty} e^k_n c_n = c_k = 0.$$

Thus, $c = 0 \in \ell^\infty_{\omega^{-1},\mathbb{N}}$.

To prove that J is surjective consider $A \in (\ell^1_{\omega,\mathbb{N}})^*$ and define $\hat{c} = \{\hat{c}_n \stackrel{\text{def}}{=} Ae^n\}_{n\in\mathbb{N}}$. We claim that $\hat{c} \in \ell^\infty_{\omega^{-1},\mathbb{N}}$ and that $J\hat{c} = A$. Observe that

$$\|\hat{c}\|_{\infty,\omega^{-1},\mathbb{N}} := \sup_{n\in\mathbb{N}} \frac{|\hat{c}_n|}{\omega_n} = \sup_{n\in\mathbb{N}} \frac{|Ae^n|}{\omega_n} \leq \sup_{n\in\mathbb{N}} \frac{\|A\| \|e^n\|_{1,\omega,\mathbb{N}}}{\omega_n} = \sup_{n\in\mathbb{N}} \frac{\|A\|\omega_n}{\omega_n} = \|A\| < \infty, \quad (7.5)$$

and hence $\hat{c} \in \ell^\infty_{\omega^{-1},\mathbb{N}}$. Furthermore, for any $a \in \ell^1_{\omega,\mathbb{N}}$

$$\begin{aligned}
(J\hat{c})a &= \sum_{n=0}^{\infty} \hat{c}_n a_n \\
&= \sum_{n=0}^{\infty} Ae^n a_n \\
&= \lim_{N\to\infty} \sum_{n=0}^{N} a_n Ae^n \\
&= \lim_{N\to\infty} A\left(\sum_{n=0}^{N} a_n e^n\right) && \text{by linearity of } l \\
&= A\left(\lim_{N\to\infty} \sum_{n=0}^{N} a_n e^k\right) && \text{by continuity of } l \\
&= A\left(\sum_{n=0}^{\infty} a_n e^n\right) \\
&= Aa.
\end{aligned}$$

Therefore, $J\hat{c} = A$.

This proves that J is a continuous linear isomorphism. Hence by Theorem 7.1.12 J^{-1} is bounded and therefore continuous.

The final step is to show that J is an isometry. Let $c \in \ell^\infty_{\omega^{-1},\mathbb{N}}$. Observe that $Q_c e^n = c_n$. Thus, $|Q_c e^n| \leq \|Q_c\| \|e^n\|_{1,\omega,\mathbb{N}} = \|Q_c\| w_n$. This implies that

$$\|c\|_{\infty,\omega^{-1},\mathbb{N}} = \sup_{n\in\mathbb{N}} \frac{|Q_c e^n|}{\omega_n} \leq \sup_{n\in\mathbb{N}} \frac{\|Q_c\|\omega_n}{\omega_n} = \|Q_c\| = \|Jc\| \leq \|J\| \|c\|_{\infty,\omega^{-1},\mathbb{N}} \leq \|c\|_{\infty,\omega^{-1},\mathbb{N}},$$

since $\|J\| \leq 1$ by (7.4). Therefore,

$$\|Jc\| = \|c\|_{\infty,\omega^{-1},\mathbb{N}},$$

which implies that J is an isometry. □

When working with Banach spaces of infinite sequences, the dual space provides a convenient means for studying more general linear operators. Let $\pi_n \colon \ell^1_{\omega,\mathbb{N}} \to \mathbb{C}$ denote the canonical projection maps; that is, given $b = \{b_n\}_{n=0}^\infty \in \ell^1_{\omega,\mathbb{N}}$, $\pi_n b = b_n$. Then a linear map $A \colon \ell^1_{\omega,\mathbb{N}} \to \ell^1_{\omega,\mathbb{N}}$ can be expressed as

$$\{(\pi_n \circ A)b\}_{n\in\mathbb{N}} = \{[Ab]_n\}_{n\in\mathbb{N}}\,.$$

If we now assume that A is bounded, then each individual linear functional $\pi_n \circ A$ is bounded, and can be expressed uniquely as

$$[Ab]_n = a_n \cdot b = \sum_{k=0}^{\infty} a_{n,k} b_k, \tag{7.6}$$

where $a_n \overset{\text{def}}{=} \{a_{n,k}\}_{k\geq 0} \in \ell^{\infty}_{\omega^{-1},\mathbb{N}}$.

Proposition 7.3.12. *Let $A\colon \ell^1_{\omega,\mathbb{N}} \to \ell^1_{\omega,\mathbb{N}}$ be a bounded linear operator. For any $n \geq 0$, consider $a_n = \{a_{n,k}\}_{k\geq 0} \in \ell^{\infty}_{\omega^{-1},\mathbb{N}}$ satisfying (7.6) which corresponds to the n^{th} row of A. Then*

$$\|A\| \leq \sum_{n=0}^{\infty} \left(\|a_n\|_{\infty,\omega^{-1},\mathbb{N}}\right) \omega_n. \tag{7.7}$$

Proof. For $b = \{b_n\}_{n\in\mathbb{N}} \in \ell^1_{\omega,\mathbb{N}}$, observe that

$$\begin{aligned}
\|Ab\|_{1,\omega,\mathbb{N}} &= \sum_{n=0}^{\infty} |[Ab]_n|\, \omega_n = \sum_{n=0}^{\infty} \left| \sum_{k=0}^{\infty} a_{n,k} b_k \right| \omega_n \\
&\leq \sum_{n=0}^{\infty} \|a_n\|_{\infty,\omega^{-1},\mathbb{N}} \|b\|_{1,\omega,\mathbb{N}} \omega_n = \|b\|_{1,\omega,\mathbb{N}} \sum_{n=0}^{\infty} \|a_n\|_{\infty,\omega^{-1},\mathbb{N}} \omega_n,
\end{aligned}$$

where the inequality follows from Lemma 7.3.11. Thus

$$\|A\| = \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \|b\|_{1,\omega,\mathbb{N}} \sum_{n=0}^{\infty} \|a_n\|_{\infty,\omega^{-1},\mathbb{N}} \omega_n = \sum_{n=0}^{\infty} \|a_n\|_{\infty,\omega^{-1},\mathbb{N}} \omega_n. \qquad \square$$

Proposition 7.3.13. *Let $A\colon \ell^1_{\omega,\mathbb{N}} \to \ell^1_{\omega,\mathbb{N}}$ be a bounded linear operator. Denote by $a_n = \{a_{m,n}\}_{m\geq 0}$ the n^{th} column of A. Let*

$$K \overset{\text{def}}{=} \sup_{n\geq 0} \frac{1}{\omega_n} \|a_n\|_{1,\omega,\mathbb{N}} = \sup_{n\geq 0} \frac{1}{\omega_n} \left(\sum_{m\geq 0} |a_{m,n}| \omega_m \right). \tag{7.8}$$

Then

$$\|A\| \leq K.$$

Proof. Given $b = \{b_n\}_{n\in\mathbb{N}} \in \ell^1_{\omega,\mathbb{N}}$,

$$\begin{aligned}
\|A\| &= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \|Ab\|_{1,\omega,\mathbb{N}} \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{m\geq 0} \left| \sum_{n\geq 0} a_{m,n} b_n \right| \omega_m \\
&\leq \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{m\geq 0} \sum_{n\geq 0} |a_{m,n}\omega_m b_n| \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{n\geq 0} \sum_{m\geq 0} |a_{m,n}\omega_m b_n| \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{n\geq 0} \left(\sum_{m\geq 0} |a_{m,n}|\omega_m \right) |b_n| \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{n\geq 0} \|a_n\|_{1,\omega,\mathbb{N}} |b_n|,
\end{aligned}$$

where the third equality follows from Fubini's theorem for infinite sums. For $n \geq 0$, let

$$\bar{c}_n \stackrel{\text{def}}{=} \|a_n\|_{1,\omega,\mathbb{N}} = \sum_{m\geq 0} |a_{m,n}|\omega_m.$$

Observe that

$$\|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}} \stackrel{\text{def}}{=} \sup_{n\in\mathbb{N}} \frac{|\bar{c}_n|}{\omega_n} = K.$$

Hence,

$$\|A\| \leq \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{n\in\mathbb{N}} |\bar{c}_n||b_n| \leq \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}} \|b\|_{1,\omega,\mathbb{N}} = \|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}}$$

where the latter inequality follows from Lemma 7.3.11. $\square$

The following Proposition is a special case of the previous result.

Proposition 7.3.14. *Consider a linear operator $A\colon \ell^1_{\omega,\mathbb{N}} \to \ell^1_{\omega,\mathbb{N}}$ of the form*

$$A = \begin{bmatrix} A_N & & 0 & \\ & c_{N+1} & & \\ & & c_{N+2} & \\ & 0 & & \ddots \end{bmatrix}$$

where $A_N = \left[a_{m,n}\right]_{m,n\in\{0,1,\dots,N\}} \in M_{N+1}(\mathbb{C})$ *and* $\{c_n\} \subset \mathbb{C}$. *If there exists a constant* C *such that* $|c_n| \le C$, *for all* $n \ge N+1$, *then*

$$\|A\| \le \max(K, C)$$

where

$$K \stackrel{\text{def}}{=} \max_{0\le n\le N} \frac{1}{\omega_n} \sum_{m=0}^{N} |a_{m,n}|\omega_m. \tag{7.9}$$

Proof. Given $b = \{b_n\}_{n\in\mathbb{N}} \in \ell^1_{\omega,\mathbb{N}}$, let $b_F = (b_0, \dots, b_N)^T$. By definition

$$\begin{aligned}
\|A\| &= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \|Ab\|_{1,\omega,\mathbb{N}} \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \left\| \begin{bmatrix} A_N & & & \\ & c_{N+1} & & \\ & & c_{N+2} & \\ & & & \ddots \end{bmatrix} \begin{bmatrix} b_F \\ b_{N+1} \\ b_{N+2} \\ \vdots \end{bmatrix} \right\|_{1,\omega,\mathbb{N}} \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \left(\sum_{m=0}^{N} \left| \sum_{n=0}^{N} a_{m,n} b_n \right| \omega_m + \sum_{n=N+1}^{\infty} |c_n b_n| \omega_n \right) \\
&= \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \left(\sum_{n=0}^{N} \left| \sum_{m=0}^{N} a_{m,n} \omega_m b_n \right| + \sum_{n=N+1}^{\infty} |c_n b_n| \omega_n \right) \\
&\le \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \left(\sum_{n=0}^{N} \left(\sum_{m=0}^{N} |a_{m,n}|\omega_m \right) |b_n| + \sum_{n\ge N+1} |c_n|\omega_n |b_n| \right).
\end{aligned} \tag{7.10}$$

For $n \ge 0$, define

$$\bar{c}_n = \begin{cases} \sum_{m=0}^{N} |a_{m,n}|\omega_m & \text{if } n \le N \\ |c_n|\omega_n & \text{if } n \ge N+1. \end{cases}$$

Observe that $\bar{c} \stackrel{\text{def}}{=} \{\bar{c}_n\}_{n\in\mathbb{N}} \in \ell^\infty_{\omega^{-1},\mathbb{N}}$ since

$$\|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}} \stackrel{\text{def}}{=} \sup_{n\in\mathbb{N}} \frac{|\bar{c}_n|}{\omega_n} = \max(K, C).$$

By (7.10)

$$\|A\| \le \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \sum_{n\in\mathbb{N}} |\bar{c}_n||b_n| \le \sup_{\|b\|_{1,\omega,\mathbb{N}}=1} \|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}} \|b\|_{1,\omega,\mathbb{N}} = \|\bar{c}\|_{\infty,\omega^{-1},\mathbb{N}}$$

where the latter inequality follows from Lemma 7.3.11. $\square$

7.4 Banach Algebra

In Example 7.3.3, to motivate our interest in ℓ^1_ν, we made use of the fact that a solution x to the scalar IVP problem

$$\dot{x} = f(x), \quad x(0) = x_0,$$

where f is a polynomial, can be expressed via its Taylor series $x(t) = \sum_{n=0}^\infty a_n t^n$. As is made clear in Chapter 8 our analysis and computations are perform using $a = \{a_n\}_{n=0}^N \in \ell^1_\nu$. With this in mind, consider an explicit example, $f(x) = x(x-1)$. Clearly we need to be able to evaluate x^2, i.e., we need to be able to take products of elements of ℓ^1_ν.

Definition 7.4.1. A *Banach algebra* is a Banach space X with a multiplication operation $\cdot \colon X \times X \to X$ that satisfies

$$x \cdot (y \cdot z) = (x \cdot y) \cdot z \tag{7.11}$$

$$(x+y)\cdot z = x \cdot z + y \cdot z, \quad x \cdot (y+z) = x \cdot y + x \cdot z, \tag{7.12}$$

$$\alpha(x \cdot y) = (\alpha x) \cdot y = x \cdot (\alpha y) \tag{7.13}$$

and

$$\|x \cdot y\| \le \|x\| \|y\|, \tag{7.14}$$

for all $x, y, z \in X$ and all scalars α. The Banach algebra is *commutative* if $x \cdot y = y \cdot x$, for all $x, y \in X$.

A *unit element* e of a Banach algebra X satisfies

$$x \cdot e = e \cdot x = x$$

for all $x \in X$. Observe that if a unit element exists, then it is unique.

Our immediate goal is to define a product for which $\ell^1_{\nu,\mathbb{I}}$ is commutative Banach algebra

Definition 7.4.2. Given two infinite sequences $a, b \in S_{\mathbb{N}}(\mathbb{C})$ their *Cauchy product* $a * b = \{(a*b)_n\}_{n\in\mathbb{N}}$ is defined as

$$(a*b)_n = \sum_{k=0}^n a_{n-k} b_k. \tag{7.15}$$

In general $a*b$ may not be a summable sequence even if both a and b are. Nevertheless we have the following result.

Theorem 7.4.3 (Mertens' Theorem). *Suppose that $\alpha, \beta \in \mathbb{C}$, that $a, b \in S_{\mathbb{N}}(\mathbb{C})$, and that*

$$\sum_{n=0}^\infty a_n = \alpha \quad \textit{and} \quad \sum_{n=0}^\infty b_n = \beta.$$

If one of the two series converges absolutely, then $a*b$ is a summable sequence of numbers. Moreover

$$\sum_{n=0}^\infty \sum_{k=0}^n a_{n-k} b_k = \alpha\beta.$$

Proof. Assume $\sum_{n=0}^{\infty} b_n$ converges absolutely. Define the partial sums

$$\alpha_N = \sum_{n=0}^{N} a_n, \quad \beta_N = \sum_{n=0}^{N} b_n, \quad \text{and} \quad \gamma_N = \sum_{n=0}^{N} (a * b)_n.$$

Note that we can rewrite γ_N as

$$\begin{aligned}
\gamma_N &= \sum_{n=0}^{N} \sum_{k=0}^{n} a_{n-k} b_k \\
&= a_0 b_0 \\
&+ a_1 b_0 + a_0 b_1 \\
&+ a_2 b_0 + a_1 b_1 + a_0 b_2 \\
&\vdots \\
&+ a_N b_0 + a_{N-1} b_1 + \ldots + a_1 b_{N-1} + a_0 b_N \\
&= b_0 \sum_{n=0}^{N} a_n + b_1 \sum_{n=0}^{N-1} a_n + \ldots + b_{N-1}(a_0 + a_1) + b_N a_0 \\
&= b_0 \alpha_N + b_1 \alpha_{N-1} + \ldots + b_N \alpha_0
\end{aligned}$$

by "summing down columns", i.e.

$$\gamma_N = \sum_{n=0}^{N} b_{N-n} \alpha_n.$$

Adding and subtracting α gives

$$\begin{aligned}
\gamma_N &= \sum_{n=0}^{N} b_{N-n}(\alpha_n - \alpha + \alpha) \\
&= \sum_{n=0}^{N} b_{N-n}(\alpha_n - \alpha) + \sum_{n=0}^{N} b_{N-n} \alpha \\
&= \sum_{n=0}^{N} b_{N-n}(\alpha_n - \alpha) + \beta_N \alpha. \qquad (7.16)
\end{aligned}$$

Now, given any $\epsilon > 0$ there is a $K_1 \in \mathbb{N}$ so that

$$|\alpha_N - \alpha| \leq \frac{1}{1 + \sum_{n=0}^{\infty} |b_n|} \frac{\epsilon}{3},$$

whenever $N \geq K_1$. This is due to the fact that $\alpha_N \to \alpha$ as $N \to \infty$ and that $\sum |b_n|$ converges absolutely. Similarly, since $\sum b_n$ converges, we have that $b_n \to 0$ as $n \to \infty$. Then there is a $K_2 \in \mathbb{N}$ so that

$$|b_n| \leq \frac{\epsilon}{3K_1 \left(1 + \sup_{0 \leq i \leq K_1 - 1} |\alpha_i - \alpha|\right)}$$

Finally, using that $\beta_N \to \beta$ as $N \to \infty$, we have that there is a $K_3 \in \mathbb{N}$ so that for all $N \geq K_3$ we have the bound,

$$|\beta_N - \beta| \leq \frac{1}{1 + |\beta|} \frac{\epsilon}{3}.$$

Choose $N \geq \max(K_1 + K_2, K_3)$. Recalling the formula of Equation (7.16) gives

$$\begin{aligned}
|\gamma_N - \alpha\beta| &= \left| \sum_{n=0}^{N} b_{N-n}(\alpha_n - \alpha) + \beta_N \alpha - \alpha\beta \right| \\
&= \sum_{n=0}^{K_1 - 1} |b_{N-n}||\alpha_n - \alpha| + \sum_{n=K_1}^{n} |b_{N-n}||\alpha_n - \alpha| + |\alpha_N - \alpha||\beta| \\
&\leq \epsilon.
\end{aligned}$$

□

Theorem 7.4.4. *Let $\omega = \{\omega_n\}_{n=0}^{\infty}$ be a sequence of weights with the property that $\omega_n \leq \omega_{n-k}\omega_k$, for all $0 \leq k \leq n$, and let $*$ denote the Cauchy product. Then $(\ell^1_{\omega,\mathbb{N}}, *)$ is a commutative Banach algebra, and in particular*

$$\|a * b\|_{1,\omega,\mathbb{N}} \leq \|a\|_{1,\omega,\mathbb{N}} \|b\|_{1,\omega,\mathbb{N}}, \tag{7.17}$$

for all $a, b \in \ell^1_{\omega,\mathbb{N}}$.

Proof. It is an exercise to check that $*$ satisfies the distributive and associative conditions of Definition 7.4.1, and to see that $*$ is a commutative product we simply note that

$$\begin{aligned}
\sum_{k=0}^{n} a_{n-k} b_k &= a_n b_0 + a_{n-1} b_1 + \ldots + a_1 b_{n-1} + a_0 b_n \\
&= a_0 b_n + a_1 b_{n-1} + \ldots + a_{n-1} b_1 + a_n b_0 \\
&= b_n a_0 + b_{n-1} a_1 + \ldots + b_1 a_{n-1} + b_0 a_n = \sum_{k=0}^{n} b_{n-k} a_k.
\end{aligned}$$

To see that $*$ satisfies the Banach algebra estimate claimed in Equation (7.17), note that for any $a, b \in \ell^1_{\omega,\mathbb{N}}$ we have

$$\begin{aligned}
\|a\|_{1,\omega,\mathbb{N}} \|b\|_{1,\omega,\mathbb{N}} &= \left(\sum_{n=0}^{\infty} |a_n| \omega_n \right) \left(\sum_{n=0}^{\infty} |b_n| \omega_m \right) \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \left(|a_{n-k}| \omega_{n-k} \right) \left(|b_k| \omega_k \right),
\end{aligned} \tag{7.18}$$

by applying Merten's theorem to the absolutely summable sequences $\{|a_n|\omega_n\}_{n=0}^{\infty}$ and $\{|b_n|\omega_n\}_{n=0}^{\infty}$.

By the hypothesized bound on the weights we have that

$$\begin{aligned}
\|a*b\|_{1,\omega,\mathbb{N}} &= \sum_{n=0}^{\infty} |(a*b)_n|\omega_n \\
&= \sum_{n=0}^{\infty} \left| \sum_{k=0}^{n} a_{n-k} b_k \right| \frac{\omega_k}{\omega_k} \omega_n \\
&\leq \sum_{n=0}^{\infty} \sum_{k=0}^{n} |a_{n-k}| \frac{\omega_n}{\omega_k} |b_k| \omega_k \\
&\leq \sum_{n=0}^{\infty} \sum_{k=0}^{n} \left(|a_{n-k}|\omega_{n-k}\right) \left(|b_k|\omega_k\right). \qquad (7.19)
\end{aligned}$$

Combining the estimates of Equations (7.18) and (7.19) yield the desired Banach algebra estimate or Equation (7.17). $\square$

Since $\nu^n = \nu^{n-k}\nu^k$ we have the following corollary.

Corollary 7.4.5. *$\ell^1_{\nu,\mathbb{N}}$ is a commutative Banach algebra.*

We have the following two sided analogue of Theorem 7.4.4, whose proof we leave as an exercise.

Theorem 7.4.6. *Given $a, b \in \ell^1_{\omega,\mathbb{Z}}$, define the discrete convolution product by $a * b = \{(a*b)_n\}_{n\in\mathbb{Z}}$ by*

$$(a*b)_n = \sum_{k\in\mathbb{Z}} a_{n-k} b_k. \qquad (7.20)$$

Let $\{\omega_n\}_{n\in\mathbb{Z}}$ be a sequence of weights with the property that $\omega_{n+k} \leq \omega_n\omega_k$. Then the pair $(\ell^1_{\omega,\mathbb{Z}}, *)$ is a commutative Banach algebra. Furthermore, the sequence e defined by

$$e_n \stackrel{\text{def}}{=} \delta_{n,0}$$

where δ is the Kronecker delta is the identify element.

We want to exploit the fact that the Cauchy product $*$ acts a multiplication on $\ell^1_{\nu,\mathbb{I}}$ to take general products of elements of $\ell^1_{\nu,\mathbb{I}}$ and thus introduce the following notation. Let $a \in \ell^1_{\nu,\mathbb{I}}$. Starting with $a^1 := a$ we inductively define

$$a^m \stackrel{\text{def}}{=} a * a^{m-1} \quad \text{for } m \in \mathbb{N}$$

with the convention that

$$a^0 \stackrel{\text{def}}{=} e,$$

the identity element of $\ell^1_{\nu,\mathbb{I}}$.

Throughout this book we make use of Newton-like operators, which are based on derivatives of functions. Fortunately, certain operations in a Banach algebra are always Fréchet differentiable.

Theorem 7.4.7. *Let X be a Banach algebra with product $\cdot$ and fix $c \in X$. Define the maps $F, G\colon X \to X$ by*

$$F(x) = c \cdot x,$$

and

$$G(x) = x \cdot x.$$

Then F and G are Fréchet differentiable on X. Moreover

$$DF(x)h = c \cdot h, \tag{7.21}$$

and

$$DG(x)h = x \cdot h + h \cdot x, \tag{7.22}$$

for all $h \in X$.

Proof. The formula for the derivative of F given in Equation (7.21) follows as soon as we observe that, for fixed $c \in X$, F is a linear map on X.

For the derivative of G start by letting $x, h \in X$, and let $A\colon X \to X$ be the map defined by

$$Ah = x \cdot h + h \cdot x.$$

Note that A is linear in h and that by (7.14) $A \in B(X)$. Moreover, we have that

$$\begin{aligned} G(x+h) - G(x) - Ah &= G(x+h) - G(x) - (x \cdot h + h \cdot x) \\ &= (x+h) \cdot (x+h) - (x \cdot x) - x \cdot h - h \cdot x \\ &= h \cdot h. \end{aligned}$$

Then for any $h \neq 0$ we have that

$$\begin{aligned} \frac{\|G(x+h) - G(x) - Ah\|}{\|h\|} &= \frac{\|h \cdot h\|}{\|h\|} \\ &\leq \frac{\|h\|\|h\|}{\|h\|} \\ &= \|h\|. \end{aligned}$$

Noting that

$$\lim_{h \to 0} \|h\| = 0,$$

(by the continuity of the norm), and that for $h \neq 0$

$$0 \leq \frac{\|G(x+h) - G(x) - Ah\|}{\|h\|} \leq \|h\|,$$

we have that

$$\lim_{h \to 0} \frac{\|G(x+h) - G(x) - Ah\|}{\|h\|} = 0,$$

by the squeeze theorem. Then

$$DG(x) = A,$$

as desired. □

Remark 7.4.8. Note that if X is a commutative Banach algebra then Theorem 7.4.7 gives that

$$D(x \cdot x)h = 2(x \cdot h),$$

which is analogous to

$$\frac{d}{dx}x^2 = 2x,$$

from Calculus of a single real variable.

7.5 Products of Banach Spaces

For the analysis of systems of differential equations we need to be able to represent vectors of functions, which leads to products of sequence spaces.

Starting on a general level, consider a finite collection of Banach spaces $V_1, \dots, V_L$. Set

$$\bigoplus_{l=1}^{L} V_l = V_1 \oplus \ldots \oplus V_N = \{(v_1, \dots, v_N) \mid v_l \in V_l \text{ for } 1 \leq l \leq L\}$$

and for $v = (v_1, \dots, v_N) \in \bigoplus_{l=1}^{L} V_n$ define

$$\|v\| := \max_{1 \leq l \leq L} \{\|v_l\|_{V_l}\}.$$

It is left to the reader to check that $\bigoplus_{l=1}^{L} V_l$ is a Banach space under the norm $\|\cdot\|$.

We are also interested in linear maps between products of Banach spaces. In particular, assume $\{V_j \mid j = 1, \dots, J\}$ and $\{W_i \mid i = 1, \dots, I\}$ are Banach spaces and that for all i and j, $A_{i,j} \colon V_j \to W_i$ is a bounded linear operator. Consider the linear operator $A \colon \bigoplus_{j=1}^{J} V_j \to \bigoplus_{i=1}^{I} W_i$ defined by

$$(Av)_i = \sum_{j=1}^{J} A_{i,j} v_j,$$

for $1 \leq i \leq I$. We leave it to the reader to verify that

$$\|A\|_{B(\bigoplus_{j=1}^J V_j, \bigoplus_{i=1}^I W_i)} = \max_{1 \leq i \leq I} \sum_{j=1}^{J} \|A_{ij}\|_{B(V_j, W_i)}.$$

In applications it is conceptually often useful to think of A as a matrix of operators

$$A = \begin{pmatrix} A_{11} & \dots & A_{1J} \\ \vdots & \ddots & \vdots \\ A_{I1} & \dots & A_{IJ} \end{pmatrix}.$$

We motivated, at the beginning of this section, the importance of Banach algebras by considering a scalar ODE and remarking that with a polynomial nonlinearity we needed to be able to take products in ℓ^1_ν. However, the focus of this book is on systems of ODEs, so consider

$$\dot{x} = f(x), \quad x \in \mathbb{R}^L$$

where $f(x) = (f_1(x), \dots, f_L(x))$ is defined in terms of polynomials and an associated solution takes the form

$$x(t) = (x_1(t), \dots, x_L(t)) = \left(\sum_{n=0}^{\infty} a_{1,n} t^n, \dots, \sum_{n=0}^{\infty} a_{L,n} t^n \right)$$

where $a = (a_1, \dots a_L) = \left(\{a_{1,n}\}_{n=0}^{\infty}, \dots, \{a_{L,n}\}_{n=0}^{\infty} \right) \in (\ell^1_\nu)^L$. This suggests that we need to be able to evaluate

$$f(x(t)) = f\left(\sum_{n=0}^{\infty} a_{1,n} t^n, \dots, \sum_{n=0}^{\infty} a_{L,n} t^n \right)$$

or, in a suggestive misuse of notation, $f(a) = f(a_1, \dots, a_L)$.

This leads us to an interjection of notation. Let $\alpha = (\alpha_1, \dots, \alpha_L) \in \mathbb{N}^L$ denote a multi-index and define $|\alpha| := \sum_{l=1}^{L} \alpha_l$. Observe that terms in the polynomial expression for f take the form

$$x^\alpha = \prod_{l=1}^{L} x_l^{\alpha_l}.$$

where $x = (x_1, \dots, x_L) \in \mathbb{R}^L$, and if M denotes the maximal degree of the polynomials $f_1, \dots, f_L$, then we can write

$$f(x) = \sum_{|\alpha|=0}^{M} c_\alpha x^\alpha \quad \text{where } c_\alpha = (c_{1,\alpha}, \dots, c_{L,\alpha}) \in \mathbb{R}^L. \tag{7.23}$$

Returning to evaluations involving the Taylor series representation of x (7.23) takes the form

$$f_l(x(t)) = \sum_{n=0}^{\infty} \phi_l(a)_n t^n, \quad l = 1, \dots, L \tag{7.24}$$

where

$$\phi_l(a) = \sum_{|\alpha|=0}^{M} c_{l,\alpha} a^{\alpha} = \sum_{|\alpha|=0}^{M} c_{l,\alpha}\, a_1^{\alpha_1} * \cdots * a_n^{\alpha_n}. \tag{7.25}$$

Observe that this expression is just the polynomial f_l with multiplication re-interpreted as the Cauchy product $*$.

We need to compute derivatives and we leave it to the reader to check that

$$D_j\phi_l(a) = \frac{\partial}{\partial a_j}\phi_l(a) = \begin{cases} \sum_{|\alpha|=0}^{M} \alpha_j c_{\alpha,l}\, a_1^{\alpha_1} * \cdots * a_j^{\alpha_j - 1} * \cdots a_n^{\alpha_n} & \text{if } \alpha_j \geq 1 \\ 0 & \text{otherwise.} \end{cases} \tag{7.26}$$

7.6 Radii Polynomial Approach on Banach spaces

Consider the problem of finding a fixed point of a nonlinear map $T\colon X \to X$ with X a Banach space, or of finding a zero of a function $F\colon X \to Y$ with X and Y Banach spaces. In the first half of the book, when considering finite dimensional problems, we found that the contraction mapping theorem and Newton's method were the appropriate tools. We even developed a-posteriori analysis which facilitated computer assisted proofs based on these ideas. In this Section we aim to extend these tools to the setting of Banach spaces.

The following theorem facilitates a-posteriori analysis of fixed point problems.

Theorem 7.6.1. *Suppose that X is a Banach space, that $T\colon X \to X$ is a Fréchet differentiable mapping, and that $\bar{x} \in X$. Let $Y_0 \geq 0$ and $Z\colon (0,\infty) \to [0,\infty)$ a non-negative function satisfying*

$$\|T(\bar{x}) - \bar{x}\|_X \leq Y_0, \tag{7.27}$$

and

$$\sup_{x \in \overline{B_r(\bar{x})}} \|DT(x)\|_{B(X)} \leq Z(r). \tag{7.28}$$

Define the radii polynomial

$$p(r) := Z(r)r - r + Y_0.$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ so that $T(\tilde{x}) = \tilde{x}$.

Proof. By the contraction mapping theorem, it is sufficient to show that $T\colon \overline{B_{r_0}(\bar{x})} \to \overline{B_{r_0}(\bar{x})}$ is a contraction mapping.

We begin by showing that T restricted to $\overline{B_{r_0}(\bar{x})}$ is a contraction mapping. Let $x_1, x_2 \in \overline{B_{r_0}(\bar{x})}$. By the Mean Value Theorem (Theorem 7.2.4),

$$\begin{aligned}\|T(x_1) - T(x_2)\|_X &\leq \sup_{x \in \overline{B_{r_0}(\bar{x})}} \|DT(x)\|_{B(X)} \|x_1 - x_2\|_X \\ &\leq Z(r_0)\|x_1 - x_2\|_X. \end{aligned} \tag{7.29}$$

Observe that

$$(Z(r_0) - 1)r_0 \leq (Z(r_0) - 1)r_0 + Y_0 = p(r_0) < 0$$

and hence $Z(r_0) < 1$. Therefore, T is a contraction mapping.

To complete the proof we show that $T\left(\overline{B_{r_0}(\bar{x})}\right) \subset \overline{B_{r_0}(\bar{x})}$. Let $y \in \overline{B_{r_0}(\bar{x})}$. Observe that

$$\begin{aligned}\|T(y) - \bar{x}\|_X &\leq \|T(y) - T(\bar{x})\|_X + \|T(\bar{x}) - \bar{x}\|_X \\ &\leq Z(r_0)\|y - \bar{x}\|_X + \|T(\bar{x}) - \bar{x}\|_X && \text{by (7.29)} \\ &\leq Z(r_0)r_0 + Y_0 && \text{by (7.27)} \\ &< r_0. && \square\end{aligned}$$

Consider a Fréchet differentiable map $F\colon X \to X$. In line with the philosophy of Chapter 2 assume that $\bar{a} \in X$ is an approximate zero of F, i.e., $\|F(\bar{a})\|_X \approx 0$, and that $DF(\bar{a})$ is invertible. Then, fixed points of the Newton-like map

$$\hat{T}(a) = a - DF(\bar{a})^{-1}F(a)$$

are in one-to-one correspondence with zeros of F. Thus, if $\hat{T}$ is a contraction mapping in a neighborhood of $\bar{a}$, then we can conclude the existence and uniqueness of a zero of F.

As is observed repeatedly in the examples of Sections 2.4 and 4.5 and Chapter 3 inverting $DF(\bar{a})$ is in some sense "too hard." Thus we make use of an approximate inverse, typically obtained by computing a numerical inverse. However, we are now working with infinite dimensional operators, thus finding an exact expression for $DF(\bar{a})^{-1}$ is even more challenging and making use of a numerical inverse is no longer an option. With this in mind we adopt a slightly different approach and approximate $DF(\bar{a})$ by an operator $A^\dagger \in B(X)$ specifically chosen to be easier to work with. We then choose $A \in B(X)$ to be an approximate inverse of $A^\dagger$ and define the Newton-like operator

$$T(a) = a - AF(a).$$

Observe that if A is injective, then fixed points of T correspond to zeros of F. The following theorem provides conditions under which we can guarantee the existence of a fixed point of T and hence a zero of F. The assumptions are more general in the sense that we do not require that $F\colon X \to X$ but instead that $AF\colon X \to X$.

Theorem 7.6.2 (Radii polynomial approach on Banach spaces). *Let X and Y be Banach spaces and $F\colon X \to Y$ be a Fréchet differentiable mapping. Suppose that $\bar{x} \in X$, $A^\dagger \in B(X,Y)$, and $A \in B(Y,X)$. Moreover assume that A is injective. Let Y_0, Z_0, and Z_1 be positive constants and $Z_2\colon (0,\infty) \to [0,\infty)$ be a non-negative function satisfying*

$$\|AF(\bar{x})\|_X \le Y_0, \tag{7.30}$$

$$\|I - AA^\dagger\|_{B(X)} \le Z_0, \tag{7.31}$$

$$\|A[DF(\bar{x}) - A^\dagger]\|_{B(X)} \le Z_1, \tag{7.32}$$

and

$$\|A[DF(c) - DF(\bar{x})]\|_{B(X)} \le Z_2(r)r, \quad \text{for all } c \in \overline{B_r(\bar{x})} \text{ and all } r > 0. \tag{7.33}$$

Define

$$p(r) := Z_2(r)r^2 - (1 - Z_0 - Z_1)r + Y_0. \tag{7.34}$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a unique $\tilde{x} \in B_{r_0}(\bar{x})$ satisfying $F(\tilde{x}) = 0$.

Proof. The idea of the proof is to define the Newton-like operator $T\colon X \to X$ by

$$T(x) = x - AF(x)$$

and to apply Theorem 7.6.1.

Since F is Fréchet differentiable and $A \in B(Y,X)$, T is Fréchet differentiable and

$$DT(x) = I - ADF(x).$$

Define $Z(r) \stackrel{\text{def}}{=} Z_0 + Z_1 + Z_2(r)r$, and observe that for $x \in \overline{B_r(\bar{x})}$,

$$\begin{aligned}\|DT(x)\|_{B(X)} &= \|I - ADF(x)\|_{B(X)} \\ &\le \|I - AA^\dagger\|_{B(X)} + \|A[A^\dagger - DF(\bar{x})]\|_{B(X)} + \|A[DF(\bar{x}) - DF(x)]\|_{B(X)} \\ &\le Z_0 + Z_1 + Z_2(r)r = Z(r) \end{aligned} \tag{7.35}$$

where the last inequality follows from assumptions (7.31), (7.32) and (7.33).

Moreover, from (7.30), $\|T(\bar{x}) - \bar{x}\|_X = \|AF(\bar{x})\|_X \le Y_0$. Since

$$p(r) = Z_2(r)r^2 - (1 - Z_0 - Z_1)r + Y_0 = Z(r)r - r + Y_0$$

and $p(r_0) < 0$, we conclude from Theorem 7.6.1 that there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ so that $T(\tilde{x}) = \tilde{x}$. Since A is assumed to be injective, $F(\tilde{x}) = 0$ if and only if $T(\tilde{x}) = \tilde{x}$. □

We conclude this section with some trivial but useful results concerning the hypothesis of Theorem 7.6.2.

Corollary 7.6.3. *Let $p(r)$ be the radii polynomial* (7.34)*. If there exists $r_0 > 0$ such that $p(r_0) < 0$, then $Z_0 + Z_1 < 1$.*

Combining Corollary 7.6.3 with Proposition 7.6.5 gives the following result.

Corollary 7.6.4. *Let $A^\dagger \in B(X,Y)$ and $A \in B(Y,X)$ be as in Theorem 7.6.2. Let $p(r)$ be the radii polynomial* (7.34)*. If there exists $r_0 > 0$ such that $p(r_0) < 0$, then $AA^\dagger$ is injective.*

Observes that this suggests that if we implement the Radii Polynomial Approach and find $r_0 > 0$ such that $p(r_0) < 0$, then perhaps it is not necessary to explicitly check that A is injective. The following result, which is applicable to all the problems considered in this text, provides conditions under which this is the case.

Proposition 7.6.5. *Consider $A \in B(Y,X)$ and $A^\dagger \in B(X,Y)$ where X and Y are Banach spaces such that $\|I - AA^\dagger\|_{B(X)} < 1$. Assume that $X = X_0 \oplus X_1$ and $Y = Y_0 \oplus Y_1$, where $X_0 \cong Y_0$ are finite dimensional. Assume that the operators A and $A^\dagger$ can be decomposed as follows*

$$A = \iota_0 \circ A_0 \circ \pi_0 + \iota_1 \circ A_1 \circ \pi_1 \tag{7.36}$$

$$A^\dagger = \iota_0 \circ A_0^\dagger \circ \pi_0 + \iota_1 \circ A_1^\dagger \circ \pi_1 \tag{7.37}$$

where for $i = 0, 1$, $A_i \colon Y_i \to X_i$, $A_i^\dagger \colon X_i \to Y_i$, and ι_i and π_i are the appropriate canonical inclusion and projection maps. Furthermore, assume that A_1 is injective. Then, A is injective.

Proof. The assumption that A_1 is injective combined with (7.36) implies that it is sufficient to prove that A_0 is injective. By Corollary 7.6.4 $AA^\dagger$ is injective and therefore by (7.36) and (7.37) $A_0 B_0 \colon X_0 \to X_0$ is injective. The assumption that X_0 is finite dimensional implies that $A_0 B_0$ is an isomorphism. The assumption that $X_0 \cong Y_0$ implies that A_0 and B_0 are isomorphisms and therefore A_0 is injective. □

7.7 A first example: Taylor series solution of parameterized equilibria

Consider the simplest case of a one-parameter family of one-dimensional ODEs,

$$\dot{x} = f(x, \lambda), \quad x \in \mathbb{R}, \lambda \in \mathbb{R}.$$

Assume $f(x_0, \lambda_0) = 0$. If $f_x(x_0, \lambda_0) \neq 0$, then the implicit function theorem implies that there exist a neighborhood U of λ_0 and a smooth function $x \colon U \to \mathbb{R}$ such that $x(\lambda_0) = x_0$ and

$$f(x(\lambda), \lambda) = 0, \quad \lambda \in U. \tag{7.38}$$

If f is analytic, then x is analytic and hence there exists $\tau > 0$ on which $x\colon (\lambda_0-\tau, \lambda_0+\tau) \to \mathbb{R}$ can be represented via the power series expansion

$$x(\lambda) = \sum_{n=0}^{\infty} a_n(\lambda - \lambda_0)^n \tag{7.39}$$

with $a_0 = x_0$.

In keeping with the philosophy of this book our goal is to provide both an explicit approximation $\bar{x}\colon U \to \mathbb{R}$ of x and an explicit bound $\|\bar{x} - x\|_\infty$ over U of the error of our approximation. The Taylor series expansion suggests how this might be done: find $\{\bar{a}_n\}_{n=0,\dots,N}$ where $\bar{a}_n \approx a_n$ such that

$$f\left(\sum_{n=0}^{N} \bar{a}_n(\lambda - \lambda_0)^n, \lambda\right) \approx 0, \quad \lambda \in U$$

and then use the Radii Polynomial approach to prove that there exists $\tilde{a} = \{\tilde{a}_n\}_{n\in\mathbb{N}}$ such that $\tilde{a} \approx \bar{a}$ and

$$f\left(\sum_{n=0}^{\infty} \tilde{a}_n(\lambda - \lambda_0)^n, \lambda\right) = 0, \quad \lambda \in U.$$

Before turning to the mathematical technicalities let us formally consider an explicit example,

$$f(x(\lambda), \lambda) = [x(\lambda)]^2 - \lambda = 0. \tag{7.40}$$

Substituting the power series expansion for x into Equation (7.40) leads to

$$\begin{aligned}
0 &= f(x(\lambda), \lambda) \\
&= \left(\sum_{n=0}^{\infty} a_n(\lambda - \lambda_0)^n\right)^2 - \lambda \\
&= \sum_{n=0}^{\infty} (a * a)_n(\lambda - \lambda_0)^n - \lambda \\
&= a_0^2 + 2a_0a_1(\lambda - \lambda_0) + \sum_{n=2}^{\infty}\sum_{j=0}^{n} a_{n-j}a_j(\lambda - \lambda_0)^n - (\lambda - \lambda_0) - \lambda_0,
\end{aligned}$$

where

$$(a * a)_n \stackrel{\text{def}}{=} \sum_{j=0}^{n} a_{n-j}a_j$$

is the Cauchy product. Matching like powers of $\lambda - \lambda_0$ leads to the infinite system of equations

$$\begin{aligned}
a_0^2 - \lambda_0 &= 0 \\
2a_0a_1 - 1 &= 0 \\
(a * a)_n &= 0, \quad n \geq 2,
\end{aligned} \tag{7.41}$$

with infinitely many unknowns $\{a_n\}_{n\in\mathbb{N}}$.

Setting $a = \{a_n\}_{n=0}^{\infty}$ we recast the problem of finding an explicit description of the curve of equilibria to that of solving

$$F(a) = 0 \tag{7.42}$$

where the nonlinear mapping F is defined component-wise by

$$F_n(a) \stackrel{\text{def}}{=} \begin{cases} a_0^2 - \lambda_0 & \text{if } n = 0 \\ 2a_1a_0 - 1 & \text{if } n = 1 \\ (a * a)_n & \text{if } n \geq 2. \end{cases} \tag{7.43}$$

Observe that the map F may be more densely written as

$$F(a) = a * a - c, \tag{7.44}$$

where $c = \{c_n\}_{n\in\mathbb{N}}$ is given by

$$c_n = \begin{cases} \lambda_0 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ 0 & \text{if } n \geq 2. \end{cases}$$

Having derived a purely formal zero finding problem (7.42), we now turn to the mathematical technicalities that puts us into a position to apply Theorem 7.6.2 to find solutions of $F(a) = 0$. We treat this as a five step process. The first four steps establish the appropriate Banach spaces, an approximate solution, and the linear operators A and $A^\dagger$. The fifth step is to establish the Y_0, Z_0, Z_1, and Z_2 bounds.

The first step towards applying Theorem 7.6.2 is to choose appropriate Banach spaces X and Y. Observe that since we are assuming that f is analytic the Taylor series (7.39) converges absolutely and uniformly for all $|\lambda - \lambda_0| < \tau$, that is

$$\sum_{n=0}^{\infty} |a_n|\nu^n < \infty \tag{7.45}$$

for all $\nu \in [0, \tau)$. This implies that $a = \{a_k\}_{k\in\mathbb{N}} \in \ell^1_{\nu,\mathbb{N}}$ and therefore we choose $X = \ell^1_{\nu,\mathbb{N}}$. By Theorem 7.4.4, X is a commutative Banach algebra. This implies that the use of the Cauchy product in (7.44) is well defined and $F\colon \ell^1_{\nu,\mathbb{N}} \to \ell^1_{\nu,\mathbb{N}}$. Therefore, we choose $Y = \ell^1_{\nu,\mathbb{N}}$. Observe that by Theorem 7.4.7 F is Fréchet differentiable.

The second step towards applying Theorem 7.6.2 is to choose $\bar{a} \in \ell^1_{\nu,\mathbb{N}}$, an approximate zero of F. From a computational perspective we cannot work with an arbitrary infinite sequence and thus, as suggested above, we choose $\bar{a}$ of the form

$$\bar{a}_k = \begin{cases} \bar{a}_k^{(N)} & \text{if } k = 0, \dots, N \\ 0 & \text{otherwise} \end{cases}$$

where $\bar{a}^{(N)} = (\bar{a}_0^{(N)}, \dots, \bar{a}_N^{(N)}) \in \mathbb{R}^{N+1}$. Observe that we can solve for $\bar{a}^{(N)}$ by solving a truncated version of (7.41)

$$\begin{aligned} a_0^2 - \lambda_0 &= 0 \\ 2a_0a_1 - 1 &= 0 \\ (a * a)_n &= 0, \quad 2 \leq n \leq N. \end{aligned} \tag{7.46}$$

For simplicity of presentation, for $n = 0, \dots, N$, we use the notation $\bar{a}_n$ as opposed to $\bar{a}_n^{(N)}$.

The third step towards applying Theorem 7.6.2 is to choose an operator $A^\dagger$ that approximates $DF(\bar{a})$. With this in mind consider the map $F^{(N)} \colon \mathbb{R}^{N+1} \to \mathbb{R}^{N+1}$ defined by

$$F_n^{(N)}(a_0, \dots, a_N) \stackrel{\text{def}}{=} \begin{cases} a_0^2 - \lambda_0 & \text{if } n = 0 \\ 2a_1a_0 - 1 & \text{if } n = 1 \\ (a * a)_n & \text{if } 2 \leq n \leq N. \end{cases}$$

Observe that we can explicitly compute $DF^{(N)}(\bar{a})$. In particular,

$$(DF(a))_{0,0} = 2\bar{a}_0,$$

for $1 \leq i \leq n$,

$$(DF(a))_{n,i} = \frac{\partial F_k}{\partial a_i} = \frac{\partial}{\partial a_i}(a * a)_n = \frac{\partial}{\partial a_i}\left(\sum_{j=0}^{n} a_{n-j}a_j\right) = 2a_{n-i},$$

and for $i > n$,

$$\frac{\partial F_n}{\partial a_i} = 0$$

since F_n only depends on the variables $a_0, \dots, a_N$.

Therefore we think of the operator $DF(\bar{a})$ as a lower triangular infinite-dimensional matrix with all the diagonal terms given by $2\bar{a}_0$. Consider the lower diagonal terms. Without computing $\bar{a}$ we do not have explicit information concerning $(DF(\bar{a}))_{n,i}$ for $0 \leq n - i \leq N$. However, $a \in \ell^1_{\nu,\mathbb{N}}$ implies that $a_n \to 0$ rapidly as $n \to \infty$. In particular, as the difference $n - i \geq 0$ grows – in other words as we move away from the diagonal in the "southwest" direction – the term $(DF(a))_{n,i} = 2a_{n-i}$ decreases rapidly to zero. Our true solution $\tilde{a} \in \ell^1_{\nu,\mathbb{N}}$, thus we can hope that for $0 \leq n - i \leq N$ the terms of $(DF(\bar{a}))_{n,i} = 2\bar{a}_{n-i}$ are not consequential.

This suggests that we approximate $DF(\bar{a})$ by the operator $A^\dagger \colon \ell^1_\nu \to \ell^1_\nu$ whose action on a vector $h \in \ell^1_\nu$ is given by

$$(A^\dagger h)_k \stackrel{\text{def}}{=} \begin{cases} [DF^{(N)}(\bar{a})h^{(N)}]_n, & 0 \leq n \leq N \\ 2\bar{a}_0 h_n, & n \geq N+1 \end{cases} \tag{7.47}$$

where $h^{(N)} = (h_0, h_1, \ldots, h_N) \in \mathbb{R}^{N+1}$ is the projection of h onto its first $(N+1)$ coordinates.

The fourth step towards applying Theorem 7.6.2 is to choose an injective linear operator A that is an approximate inverse of $A^\dagger$. Assuming $a_0 \neq 0$ define $A \colon \ell^1_\nu \to \ell^1_\nu$ by

$$(Ah)_n \stackrel{\text{def}}{=} \begin{cases} [A^{(N)}h^{(N)}]_n, & 0 \leq n \leq N \\ \\ \frac{1}{2\bar{a}_0} h_n, & n \geq N+1, \end{cases} \tag{7.48}$$

where $A^{(N)}$ is a numerical inverse of $DF^{(N)}(\bar{a})$.

Conceptually, it may be of use to think of $A^\dagger$ and A as infinite square matrices

$$A^\dagger = \begin{bmatrix} DF^{(N)}(\bar{a}) & 0 \\ 0 & \Lambda \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} A^{(N)} & 0 \\ 0 & \Lambda^{-1} \end{bmatrix}$$

where Λ is an infinite diagonal matrix with constant diagonal entries $2\bar{a}_0$. A direct application of Proposition 7.6.5 implies that if we identify a value r_0 at which the radii polynomial $p(r_0) < 0$, then A is injective.

Having defined the Banach spaces $X = Y = \ell^1_\nu$, the nonlinear function $F \colon \ell^1_\nu \to \ell^1_\nu$, the linear operators $A, A^\dagger \in B(X)$, and determining that A is invertible, we turn to fifth step towards applying Theorem 7.6.2 that of identifying the Radii Polynomial bounds Y_0, Z_0, Z_1, and Z_2. These are presented in the following theorem.

Theorem 7.7.1. *Fix $\nu > 0$ and define the constants*

$$\begin{aligned} Y_0 &\stackrel{\text{def}}{=} \sum_{n=0}^{N} |[A^{(N)}F^{(N)}(\bar{a})]_n|\nu^n + \frac{1}{2|\bar{a}_0|} \sum_{n=N+1}^{2N} \sum_{j=0}^{2N-n} |\bar{a}_{N-j}||\bar{a}_{n-N+j}|\nu^n \\ Z_0 &\stackrel{\text{def}}{=} \left\| I - A^{(N)}DF^{(N)}(\bar{a}) \right\|_{1,\nu} \\ Z_1 &\stackrel{\text{def}}{=} \frac{1}{|\bar{a}_0|} \sum_{n=1}^{N} |\bar{a}_n|\nu^n \\ Z_2 &\stackrel{\text{def}}{=} 2\max\left(\|A^{(N)}\|_{1,\nu}, \frac{1}{2|\bar{a}_0|} \right) \end{aligned}$$

where, given $B \in M_{N+1}(\mathbb{R})$,

$$\|B\|_{1,\nu} \stackrel{\text{def}}{=} \max_{0 \leq n \leq N} \frac{1}{\nu^n} \sum_{m=0}^{N} |B_{m,n}|\, \nu^m.$$

Then, Y_0, Z_0, Z_1 and Z_2 as defined above satisfy the hypotheses of Theorem 7.6.2.

Proof. First note that

$$(\bar{a} * \bar{a})_n = \begin{cases} \displaystyle\sum_{j=0}^{n} \bar{a}_{n-j}\bar{a}_j, & 0 \le n \le N \\ \displaystyle\sum_{j=0}^{2N-n} \bar{a}_{N-j}\bar{a}_{n-N+j}, & N+1 \le n \le 2N \\ 0, & n \ge 2N+1 \end{cases}$$

as $\bar{a}_n = 0$ for $n \ge N+1$. Thus,

$$[AF(\bar{a})]_n = \begin{cases} [A^{(N)}F^{(N)}(\bar{a})]_n, & 0 \le n \le N \\ \displaystyle\frac{1}{2\bar{a}_0}\sum_{j=0}^{2N-n} \bar{a}_{N-j}\bar{a}_{n-N+j}, & N+1 \le n \le 2N \\ 0, & n \ge N+1 \end{cases}$$

and

$$\begin{aligned} \|AF(\bar{a})\|_{1,\nu} &= \sum_{n=0}^{\infty} |[AF(\bar{a})]_n|\nu^n \\ &= \sum_{n=0}^{N} |[A^{(N)}F^{(N)}(\bar{a})]_n|\nu^n + \sum_{n=N+1}^{2N} |[AF(\bar{a})]_n|\nu^n \\ &\le \sum_{n=0}^{N} |[A^{(N)}F^{(N)}(\bar{a})]_n|\nu^n + \frac{1}{2|\bar{a}_0|}\sum_{n=N+1}^{2N}\sum_{j=0}^{2N-n} |\bar{a}_{N-j}||\bar{a}_{n-N+j}|\nu^n = Y_0. \end{aligned}$$

Next, for $h \in \ell^1_\nu$,

$$\left[\left(I - AA^\dagger\right)h\right]_n = \begin{cases} \left[\left(I - A^{(N)}DF^{(N)}(\bar{a})\right)h^{(N)}\right]_n, & 0 \le n \le N \\ 0, & n \ge N+1. \end{cases}$$

Then

$$\|I - AA^\dagger\| = \sup_{\|h\|_{1,\nu}=1} \|[I - AA^\dagger]h\|_{1,\nu} \le \|I - A^{(N)}DF^{(N)}(\bar{a})\|_{1,\nu} = Z_0.$$

By Remark 7.4.8,

$$DF(a)h = 2a * h,$$

for $h \in \ell^1_\nu$. Moreover,

$$\begin{aligned} [(DF(\bar{a}) - A^\dagger)h]_n &= \begin{cases} [DF^{(N)}(\bar{a})h^{(N)}]_n - [DF^{(N)}(\bar{a})h^{(N)}]_n, & 0 \le n \le N \\ 2(\bar{a} * h)_n - 2\bar{a}_0 h_n, & n \ge N+1 \end{cases} \\ &= \begin{cases} 0, & 0 \le n \le N \\ 2\displaystyle\sum_{j=1}^{N} h_{n-j}\bar{a}_j, & n \ge N+1, \end{cases} \end{aligned}$$

which exploits the fact that $\bar{a}_n = 0$ for $n \geq N+1$. Then

$$[A(DF(\bar{a}) - A^{\dagger})h]_n = \begin{cases} 0, & 0 \leq n \leq N \\ \dfrac{1}{\bar{a}_0} \displaystyle\sum_{j=1}^{N} h_{n-j}\bar{a}_j, & n \geq N+1. \end{cases}$$

In order to better understand this term we define $\hat{a} \in \ell^1_\nu$ by $\hat{a}_n = 0$ if $n = 0$, or $n \geq N+1$ and $\hat{a}_n = \bar{a}^{(N)}_n$ for $1 \leq n \leq N$, i.e. $\hat{a} = (0, \bar{a}_1, \ldots, \bar{a}_N, 0, \ldots)$. Now for any $h \in \ell^1_\nu$ with $\|h\|_{1,\nu} = 1$,

$$\begin{aligned} \|A[DF(\bar{a}) - A^{\dagger}]h\|_{1,\nu} &= \sum_{n=N+1}^{\infty} \frac{1}{|\bar{a}_0|} \left| \sum_{j=1}^{N} h_{n-j}\bar{a}_j \right| \nu^n \\ &\leq \frac{1}{|\bar{a}_0|} \sum_{n=N+1}^{\infty} \left| \sum_{j=0}^{n} h_{n-j}\bar{a}_j \right| \nu^n \\ &\leq \frac{1}{|\bar{a}_0|} \sum_{n=0}^{\infty} \left| \sum_{j=0}^{n} h_{n-j}\bar{a}_j \right| \nu^n \\ &= \frac{1}{|\bar{a}_0|} \|h * \bar{a}\|_{1,\nu} \\ &\leq \frac{1}{|\bar{a}_0|} \|h\|_{1,\nu} \|\bar{a}\|_{1,\nu} \\ &\leq \frac{1}{|\bar{a}_0|} \sum_{n=1}^{N} |\bar{a}_n| \nu^n = Z_1. \end{aligned}$$

Hence, $\|A[DF(\bar{a}) - A^{\dagger}]\| \leq Z_1$.

Since $DF(a)h = 2a * h$, then

$$\|A[DF(c) - DF(\bar{x})]\| \leq 2\|A\|\|c - \bar{x}\|_{1,\nu} \leq 2\|A\|r. \tag{7.49}$$

Now, since A defined by (7.48) has the form

$$A = \begin{bmatrix} A^{(N)} & & 0 & \\ & \frac{1}{2\bar{a}_0} & & \\ & & \frac{1}{2\bar{a}_0} & \\ & 0 & & \ddots \end{bmatrix},$$

then by Proposition 7.3.14, $\|A\| \leq \max\left(\|A^{(N)}\|_{1,\nu}, \frac{1}{2|\bar{a}_0|}\right)$. From (7.49), we set

$$Z_2 = 2\max\left(\|A^{(N)}\|_{1,\nu}, \frac{1}{2|\bar{a}_0|}\right). \qquad \square$$

Consider Equation (7.40) with $\lambda_0 = 1/3$ truncated at $N = 2$. This leads to the system quadratic of equations

$$
\begin{aligned}
a_0^2 - 1/3 &= 0 \\
2a_0 a_1 - 1 &= 0 \\
a_2 a_0 + a_1 a_1 + a_0 a_2 &= 0.
\end{aligned}
$$

An approximate solution is give by

$$\bar{a} = \begin{pmatrix} \bar{a}_0 \\ \bar{a}_1 \\ \bar{a}_2 \end{pmatrix} = \begin{pmatrix} 0.57735026918962 \\ 0.86602540378443 \\ -0.64951905283832 \end{pmatrix}.$$

Moreover the numerical matrix

$$A^{(N)} = \begin{pmatrix} 0.86602540378443 & -0.00000000000000 & 0.00000000000000 \\ -1.29903810567665 & 0.86602540378443 & -0.00000000000000 \\ 2.92283573777248 & -1.29903810567665 & 0.86602540378443 \end{pmatrix}$$

approximately inverts the matrix $DF^{(N)}(\bar{a})$.

We choose $\nu = 1/4$ and check that

$$Y_0 \leq 0.016650268688483,$$

$$\|I - A^{(N)} DF^{(N)}(\bar{a})\| \leq 1.504019729180741 \times 10^{-15} =: Z_0,$$

$$Z_1 \leq 0.44531250,$$

and

$$Z_2 \leq 2.746924327628767$$

Similarly we check that if

$$0.03668029648410 = r_- \leq r \leq r_+ = 0.16525009323246,$$

the

$$p(r) \leq 0.$$

It follows that the approximate solution

$$x^{(N)}(\lambda) = \bar{a}_2(\lambda - 1/3)^2 + \bar{a}_1(\lambda - 1/3) + \bar{a}_0,$$

satisfies

$$\sup_{|\lambda| \leq 1/4} |x^{(N)}(\lambda) - \tilde{x}(\lambda)| \leq 0.03668029648410,$$

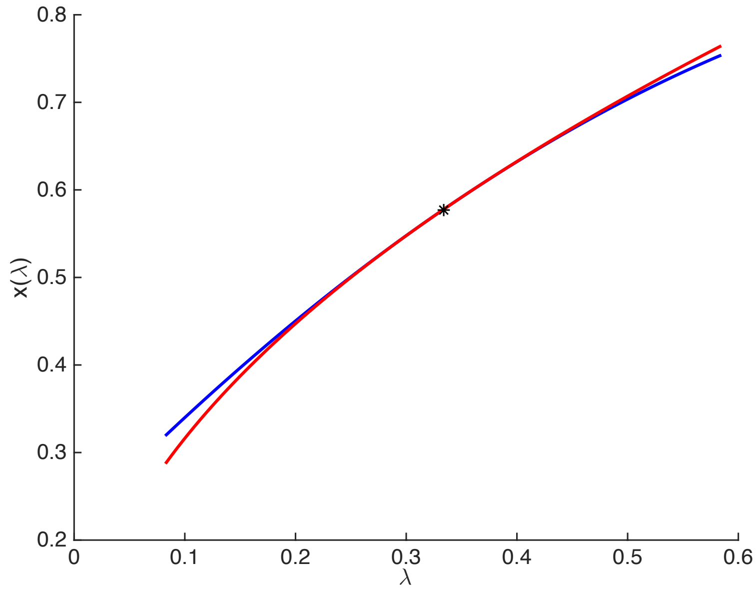

Figure 7.1: Plot of $\bar{x}^{(N)}(\lambda)$ for $N = 2$ show as a blue curve versus and $\tilde{x}(\lambda) = \sqrt{\lambda}$ shown in red. The point $(\lambda_0, x_0) = (1/3, \sqrt{1/3})$ is the black star. The approximation error is rigorously proven to be less than 0.037 in the indicated domain.

where $\tilde{x}$ is the unique function satisfying

$$\tilde{x}(\lambda)^2 - \lambda = 0, \qquad x(1/3)^2 = 1/3.$$

The example just discussed could be worked more or less by hand. Of course these results can be improved by increasing N and adjusting ν. Results of a number of additional computations are illustrated in Table 7.2.

Remark 7.7.2. In this case we can compare with the true solution. Figure 8.1 illustrates the graph of $x^{(N)}(\lambda)$ (with $N = 2$ and $\lambda_0 = 1/3$) versus the known true solution $\tilde{x}(\lambda) = \sqrt{\lambda}$. Indeed the explicit formula for the Taylor coefficients is given by

$$\tilde{a}_n = \frac{\sqrt{\lambda_0}}{\lambda_0^n} \frac{(-1)^n (2n)!}{(1-2n)(n!)^2 4^n}$$

The formula above provides valuable information about the precise decay rate of the power series coefficients. However for computational purposes it is difficult to argue that such complicated exact formulas are preferable to the validated Newton-like argument discussed above. Especially when we stop to consider that the "exact formula" for $\tilde{a}_n$ still involves $\sqrt{\lambda_0}$, a quantity which must itself be computed via a rigorously validated iterative argument.

λ_0	N	ν	r	$\lvert\bar{a}_N\rvert$
1/3	10	0.25	5.91×10^{-4}	3.1×10^{2}
1/3	18	0.25	2.9×10^{-5}	8.5×10^{5}
1/3	5	0.0025	2.22×10^{-15}	3.9
1/3	12	0.05	8.2×10^{-14}	2.15×10^{3}
0.1	10	0.01	3.0×10^{-14}	3×10^{7}
2	30	1	2.1×10^{-12}	2.3×10^{-12}
2	40	1	1.8×10^{-15}	1.5×10^{-15}
7	30	2.5	5×10^{-16}	3×10^{-28}

Figure 7.2: The table records the results of a number of additional computer-assisted proofs. Note that when $\lambda_0 < 1$ the coefficients of the power series grow. This has to be balanced by taking the domain ν smaller. It also makes high order computations numerically unstable. We achieve the best results when N is not too large. On the other hand when $\lambda_0 > 1$ the coefficients of the series decay. This stabilizes high order computations and allows us to take both N and ν larger.

7.8 Exercises

Exercise 7.8.1. Show that $C^1([a,b],\mathbb{R})$ with the norm

$$\|f\|_{C^1([a,b])} = \|f\|_{C^0([a,b])} + \|f'\|_{C^0([a,b])}.$$

is a Banach space.

Exercise 7.8.2. Let X and Y are vector spaces, and let $A : Y \to X$ and $B : X \to Y$ be a linear operators. Assume that AB is invertible. Give an example where A and B are not invertible.

Exercise 7.8.3. Let X and Y be Banach spaces and $T\colon X \to Y$ be a bijective linear map. Prove that the inverse of T is also linear.

Exercise 7.8.4. Let X and Y be normed linear spaces and $T\colon X \to Y$ be a linear operator. Prove that the sets

$$\text{image}(T) = \{y \in Y : y = T(x) \text{ for some } x \in X\},$$

and

$$\ker(T) = \{x \in X : T(x) = 0\},$$

are normed linear (sub)spaces. Prove that T is one-to-one if and only if $\ker(T) = 0$.

Prove that if X and Y are Banach spaces and $T \in B(X,Y)$, then $\ker(T)$ is a Banach space.

Exercise 7.8.5. Let ω be a one or two sided sequence of weights (depending on the context).

1. Prove that the one sided weighted supremum $\|a\|_{\infty,\omega,\mathbb{N}} := \sup_{n\geq 0} |a_n|\omega_n$ defines a norm on $S_{\mathbb{N}}(\mathbb{C})$ and that $\ell_{\infty,\omega,\mathbb{N}}$ is a Banach space.
2. Prove that the two sided weighted sum $\|a\|_{1,\omega,\mathbb{N}} := \sum_{n\in\mathbb{Z}} |a_n|\omega_n$ defines a norm on $S_{\mathbb{Z}}(\mathbb{C})$ and that $\ell_{1,\omega,\mathbb{Z}}$ is a Banach space.
3. Prove that the two sided weighted supremum $\|a\|_{\infty,\omega,\mathbb{N}} := \sup_{n\in\mathbb{Z}} |a_n|\omega_n$ defines a norm on $S_{\mathbb{Z}}(\mathbb{C})$ and that $\ell_{\infty,\omega,\mathbb{Z}}$ is a Banach space.

Exercise 7.8.6. Recall the definition of the standard two sided basis vectors from Remark 6.2.12. Prove that these vectors provide a Schauder basis for $\ell_{1,\omega,\mathbb{Z}}$.

Exercise 7.8.7. Let X and Y be Banach spaces and $A \in B(X,Y)$. Assume that $\{\phi_n\}_{n=0}^{\infty} \subset X$ is a Schauder basis for X. Prove that if $x = \sum_{n=0}^{\infty} a_n\phi_n$, then

$$Ax = \sum_{n=0}^{\infty} a_n A(\phi_n).$$

Exercise 7.8.8. Let $\omega = \{\omega_n\}_{n\in\mathbb{Z}}$ be a two sided sequence of weights. Define $\omega^{-1} := \{\omega_n^{-1}\}_{n\in Z}$. Prove that $\ell_{\infty,\omega^{-1},\mathbb{Z}}$ is isometrically isomorphic to $(\ell_{1,\omega,\mathbb{Z}})^*$.

Exercise 7.8.9. Let X and Y be Banach spaces. Assume that $F\colon X \to Y$ is Fréchet differentiable at $x_0 \in X$. Prove the following results.

1. F is continuous at x_0.
2. $DF(x_0)$ is unique.

Exercise 7.8.10. Let X and Y be Banach spaces and let $A \in B(X,Y)$. Assume that $F, G\colon X \to Y$ are Fréchet differentiable at x_0. Prove the following results.

1. Prove that A is Fréchet differentiable for all $x \in X$ and that $DA(x) = A$ i.e. for all $h \in X$,
$$DA(x)h = Ah.$$
2. Prove that AF is Fréchet differentiable at x_0 and that
$$D(AF)(x_0) = ADF(x_0).$$
3. Let $\alpha, \beta \in \mathbb{C}$. Prove that $\alpha F + \beta G$ is Fréchet differentiable at x_0 and that
$$D(\alpha F + \beta G)(x_0) = \alpha DF(x_0) + \beta DG(x_0).$$

Exercise 7.8.11. Let X, Y, and Z be Banach spaces. Assume that $F\colon X \to Y$ and $G\colon Y \to Z$ are Fréchet differentiable at $x \in X$ and $y = F(x) \in Y$, respectively. Prove that $G \circ F\colon X \to Z$ is Fréchet differentiable at x and that

$$D(G \circ F)(x) = DG(y)DF(x).$$

Exercise 7.8.12. Let X be a commutative Banach algebra with product $*$ and let $p \geq 3$ be an integer. Define $G\colon X \to X$ by $G(x) = x^p$. Then G is Fréchet differentiable and

$$DG(x)h = px^{p-1} * h, \tag{7.50}$$

for all $h \in X$.

Exercise 7.8.13. Let $f \in C^1(\mathbb{R}^2, \mathbb{R})$. Consider $F\colon C^0([a,b],\mathbb{R}) \to C^0([a,b],\mathbb{R})$ defined by

$$F(x)(\lambda) := f(\lambda, x(\lambda)),$$

for $\lambda \in [a,b]$ and $x \in C^0([a,b],\mathbb{R})$. Prove that the Fréchet derivative of F is given by

$$DF(x)h(\lambda) = \frac{\partial f}{\partial y}(\lambda, x(\lambda))h(\lambda).$$

Exercise 7.8.14. Let U be an open subset of $\mathbb{R}^N$, $x_0 \in U$, and $f \in C^1(U, \mathbb{R}^N)$. Choose $\epsilon > 0$ such that $B_\epsilon(x_0) \subset U$. Choose M and K such that

$$\sup_{x \in B_\epsilon(x_0)} \|f(x)\| \leq M \quad \text{and} \quad \sup_{x \in B_\epsilon(x_0)} \|Df(x)\| \leq K,$$

and let $0 < a < \min\left\{\epsilon M^{-1}, K^{-1}\right\}$.

Set $X := \overline{B_\epsilon(x_0)} \subset C^0([-a,a],\mathbb{R}^N)$ where $x_0\colon [-a,a] \to \mathbb{R}^N$ denotes the constant function taking the value x_0. Define $F\colon X \to C^0([-a,a],\mathbb{R}^N)$ by

$$F(\alpha)(t) := \alpha(t) - x_0 - \int_0^t f(\alpha(s))\,ds.$$

Prove that F is Fréchet differentiable and $DF = I - B$ where B is given by

$$[Bh](t) = \int_0^t Df(\alpha(s))h(s)\,ds.$$

Furthermore, show that for $\alpha \in X$, $DF(\alpha)$ is invertible and

$$\left\|DF(\alpha)^{-1}\right\| \leq \frac{1}{1 - aK}.$$

Hint: Use the Neumann series to prove the invertibility of $DF(\alpha)$.

Exercise 7.8.15. Let X be a Banach algebra with product $*$. Prove that $*$ is a continuous operation. In particular, show that if $x_n \to x$ and $y_n \to y$, then $x_n * y_n \to x * y$.

Exercise 7.8.16. Define a product $*\colon L^1(\mathbb{R}) \times L^1(\mathbb{R}) \to L^1(\mathbb{R})$ by convolution, i.e. for $f, g \in L^1(\mathbb{R})$ let

$$(f * g)(t) = \int_{\mathbb{R}} f(x)g(t-x)\,dx,$$

be their product. Then $(L^1(\mathbb{R})), *)$ is a commutative Banach algebra.

Exercise 7.8.17. Let $\omega = \{\omega_n\}_{n\in\mathbb{Z}}$ be a sequence with

$$\frac{\omega_n}{\omega_k} \leq \omega_{n-k},$$

for all $k, n \in \mathbb{Z}$. Given $a, b \in \ell^1_{\omega,\mathbb{Z}}$ define the discrete convolution product of a and b to be the new sequence $a * b = \{(a * b)_n\}_{n\in\mathbb{Z}}$ given by

$$(a * b)_n = \sum_{k\in\mathbb{Z}} a_{n-k} b_k.$$

1. Prove that $*\colon \ell^1_{\omega,\mathbb{Z}} \times \ell^1_{\omega,\mathbb{Z}} \to \ell^1_{\omega,\mathbb{Z}}$ is a well defined binary product (the challenge here is to show that the result $a * b$ is actually an element of $\ell^1_{\omega,\mathbb{Z}}$). Show that the product commutes.

2. Prove that
$$\|a * b\|_{1,\omega,\mathbb{Z}} \leq \|a\|_{1,\omega,\mathbb{Z}} \|b\|_{1,\omega,\mathbb{Z}},$$
so that $(\ell^1_{\omega,\mathbb{Z}}, *)$ is a commutative Banach algebra.

Exercise 7.8.18. Let X be a normed linear space and Y be a Banach space. For $T\colon X \to Y$ recall that the operator define norm is defined by

$$\|T\|_{B(X,Y)} := \sup_{\|x\|_X = 1} \|T(x)\|_Y,$$

and that the set

$$B(X, Y) := \left\{T\colon X \to Y \,|\, T \text{ is linear and } \|T\|_{B(X,Y)} < \infty\right\},$$

is a Banach space. Let $T, S \in B(X, Y)$ and define the product $T * S$ by composition, i.e.

$$(T * S)(x) = T(S(x)),$$

for any $x \in X$. Prove that this product endows $B(X, Y)$ with a Banach algebra structure. What can you say about calculus in this Banach algebra?

`http://www.cnd.mcgill.ca/~ivan/PerEnfloAcounterexampleToTheApproximationProblemInBanachSpaces.pdf`

The next five exercise assume the knowledge of measure theory and the Lebesgue integral.

Exercise 7.8.19. By Example 7.1.2 $C^0([a, b])$ is not a Banach space when endowed with the norm

$$\|f\|_1 = \int_a^b |f(t)|\, dt.$$

The set

$$L^1([a,b]) = \{f \colon [a,b] \to \mathbb{R} \mid f \text{ is a measurable function, and } \|f\|_1 < \infty\},$$

is the completion of $C^0([a,b])$ under the $\|\cdot\|_1$ norm. Show that under this norm, $L^1([a,b])$ is a Banach space.

Exercise 7.8.20. Let $\Omega \subset \mathbb{R}^n$ be a measurable set and define

$$\|f\|_{1,\Omega} := \int_\Omega |f(x)|\, dx,$$

where the integral is taken in the sense of Lebesgue. Then

$$L^1(\Omega) = \{f \colon \Omega \to \mathbb{R}^n | f \text{ is measurable, and } \|f\|_{1,\Omega} < \infty\},$$

is a Banach space.

Exercise 7.8.21. Choose a fixed $g \in L^1(\mathbb{R})$ and define the map $A \colon L^1(\mathbb{R}) \to L^1(\mathbb{R})$ by

$$(Af)(t) = (f * g)(t),$$

where $f \in L^1(\mathbb{R})$.

- Prove that A is a well defined linear operator.
- Prove that A is a bounded linear operator and provide a bound on the norm.

Exercise 7.8.22. Fix $g \in L^1(\mathbb{R})$, and recall that for any other $f \in L^1(\mathbb{R})$ the *convolution* of f and g is defined by

$$(f * g)(t) := \int_{\mathbb{R}} f(x)g(t-x)\, dx.$$

Define the linear operator $A \colon L^1(\mathbb{R}) \to L^1(\mathbb{R})$ by

$$(Af)(t) = (f * g)(t).$$

Show that A is a bounded linear operator.

Exercise 7.8.23. Consider the function $N \colon L^1(\mathbb{R}) \to L^1(\mathbb{R})$ defined by

$$N[f](t) := (f * f)(t) = \int_{\mathbb{R}} f(s)f(t-s)\, ds,$$

for $f \in L^1(\mathbb{R})$.

1. Prove that N is Fréchet differentiable for each $f \in L^1(\mathbb{R})$ and compute the derivative (don't use any theorems about Banach algebras. Instead, work directly from the definition).

2. Consider the "triple" convolution

$$T[f](t) = (f * (f * f))(t).$$

   Explain why T is well defined. Moreover, show that

$$((f * f) * f)(t) = (f * (f * f))(t).$$

   Because of this associativity it makes sense to drop the parenthesis and write $T(f)(t) = (f * f * f)(t)$.

3. Prove that $N[f](t)$ is Fréchet differentiable for all $f \in L^1(\mathbb{R})$, and compute the derivative.

4. Extend the previous result by induction to any number of "monomial" convolutions of f with itself, i.e. define

$$(f^n)(t) = (f * \ldots * f)(t), \qquad \text{(convolution } n \text{ times)}.$$

   Prove that the repeated convolution makes sense (associavitity), that the result is Fréchet differentiable, and that compute the derivative.

5. Suppose that $g \in L^1(\mathbb{R})$ is fixed. Give a short explanation (one or two sentences) for why the map

$$(Af)(t) = (g * f)(t),$$

   is Fréchet differentiable. What is the derivative?

6. Let $g_0, g_1, \ldots, g_n \in L^1(\mathbb{R})$ be a collection of Lebesgue integrable functions. Define the "polynomial" operator $P\colon L^1(\mathbb{R}) \to L^1(\mathbb{R})$ by

$$P[f](t) = (g_n * f^n)(t) + \ldots + (g_2 * f^2)(t) + (g_1 * f)(t) + g_0(t).$$

   where $f \in L^1(\mathbb{R})$. Prove that P is Fréchet differentiable and compute the derivative.

7. How much of the preceding work generalizes if you work instead with a commutative Banach algebra $(X, *)$?